

## $H^\infty$ Control.

J. B. Burl. Linear Optimal Control.  
Addison Wesley, 1999, pp. 329-364.

# 9

# Full Information Control and Estimation

The linear quadratic regulator, Kalman filter, and linear quadratic Gaussian problems can all be posed as 2-norm optimization problems. These optimization problems can also be posed using the system  $\infty$ -norm as a cost function. The  $\infty$ -norm is the worst-case gain of the system and therefore provides a good match to engineering specifications, which are typically given in terms of bounds on errors and controls.

The small-gain theorem states that, for unstructured perturbations, robust stability depends on the  $\infty$ -norm of the closed-loop system from the perturbation input to the perturbation output. The minimization of the closed-loop  $\infty$ -norm can therefore also be used as a means of maximizing robustness.

Robust stability in the presence of structured perturbations, and robust performance both depend on the supremum (over frequency) of the structured singular value from the perturbation input to the perturbation output. Note that a performance block is added to the perturbation (forming a structured perturbation) for robust performance analysis. The supremum of the SSV is bounded by the  $\infty$ -norm of the diagonal-scaled, closed-loop system. In fact, this bound is typically used in place of the SSV in robustness analysis. Therefore, it is reasonable to assume that minimization of the system  $\infty$ -norm plays a role in maximizing both robust stability (for structured perturbations) and robust performance (in general).

In summary, the optimization of the  $\infty$ -norm has applications in both maximizing performance and robustness. Control and estimation problems, with the goal of minimizing of the system  $\infty$ -norm, are termed  $\mathcal{H}_\infty$  optimizations problems.<sup>1</sup> This chapter develops some basic results on the design of systems that minimize the system  $\infty$ -norm, in particular, full information control and output estimation. Full information control and output estimation are combined in the next chapter to form  $\mathcal{H}_\infty$  output feedback. The results in this chapter are derived solely within a performance optimization framework. The maximization of performance robustness is discussed in the next chapter.

## 9.1 Differential Games

An  $\mathcal{H}_\infty$  optimal controller minimizes the worst-case gain of the system. This problem can be thought of as a game with two participants: the designer, who is seeking a control that minimizes the gain; and nature, which is seeking a disturbance that maximizes the

<sup>1</sup>The  $\mathcal{H}$  in  $\mathcal{H}_\infty$  refers the Hardy space of all stable systems, and the  $\infty$  refers to the system norm. While stability is not an issue in finite-time problems, these problems are also referred to as  $\mathcal{H}_\infty$  optimization problems in keeping with standard terminology in the literature.

gain (the worst-case input). Games of this type are termed differential games when the dynamics of the game are described by differential equations. Some fundamental results from the field of differential game theory are presented in this section for application to  $\mathcal{H}_\infty$  optimization problems.

A differential game is described by the game dynamics and the objective function.<sup>2</sup> The game dynamics are modeled by a generic state equation:

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}_u u(t) + \mathbf{B}_w w(t), \quad (9.1)$$

where  $u(t)$  is the control input, which is selected by the designer, and  $w(t)$  is the disturbance input, which is selected by nature. The objective function is a real function (not necessarily positive definite) of the state, the control, and the disturbance:

$$J\{x(t), u(t), w(t)\}.$$

The solution of the differential game consists of the optimal control trajectory  $u^*(t)$  and the worst-case disturbance input  $w^*(t)$ . This solution is necessarily a saddle point of the objective function, which is defined by the following inequalities:

$$J\{x(u^*, w), u^*(t), w(t)\} \leq J\{x(u^*, w^*), u^*(t), w^*(t)\} \leq J\{x(u, w^*), u(t), w^*(t)\}.$$

The state is shown as a function of the inputs in this expression to emphasize that the state is completely specified by the inputs (provided the state initial condition is zero). A saddle point can also be defined as the argument of the mini-max problem:

$$\min_u (\max_w J\{x(u, v), u(t), w(t)\}).$$

Lagrange multipliers can be used to convert a constrained mini-max problem into an unconstrained mini-max problem of higher order. Appending the constraint equation (9.1) to the objective function yields the augmented objective function:

$$J_a(x, u, w, p) = J(x, u, w) + \int_0^{t_f} p^T(t) \{\mathbf{A}x(t) + \mathbf{B}_u u(t) + \mathbf{B}_w w(t) - \dot{x}(t)\} dt.$$

A necessary condition for a saddle point is that the variation of this augmented objective function (with respect to perturbations in the state, the control, the disturbance, and the Lagrange multiplier) must be zero. This result can be intuitively justified by noting that the mini-max problem consists of two optimization problems, or by following the procedure given in Subsection 6.1.2.

## 9.2 Full Information Control

An optimal full information controller minimizes the infinity norm of the closed-loop system from the disturbance input to the reference output, assuming that all of the plant states and all of the disturbance inputs are available for feedback. In the stochastic regulator and in LQG control, the disturbance input is white noise. This disturbance input has an infinite variance, is totally unpredictable, and is not measurable. Using this in-

<sup>2</sup> Constraints on the controls and disturbances are typically included in a differential game description. These constraints are not included in this presentation, since they are not used in the following treatment of  $\mathcal{H}_\infty$  optimization.



put for feedback is then not practical. In  $\mathcal{H}_\infty$  optimization problems, the disturbance input may be measured or predicted and is included as a possible feedback input. The term *full information* is used to reflect this change with respect to the state feedback controller, which is the solution of the linear quadratic regulator problem.

The  $\mathcal{H}_\infty$  full information control problem is formally stated below. Let the plant be described by the following state model:

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}_u u(t) + \mathbf{B}_w w(t); \quad (9.2a)$$

$$y(t) = \mathbf{C}_y x(t) + \mathbf{D}_{yu} u(t), \quad (9.2b)$$

where

$$\mathbf{D}_{yu}^T \mathbf{C}_y = \mathbf{0}; \quad (9.2c)$$

$$\mathbf{D}_{yu}^T \mathbf{D}_{yu} = \mathbf{I}. \quad (9.2d)$$

The plant is assumed to be observable from the reference output  $y(t)$  and controllable from the control input  $u(t)$ .<sup>3</sup> The zero in condition (9.2c) specifies that the output consists of two distinct components: linear combinations of the state and linear combinations of the control. This separation between the state and control is equivalent to the absence of cross terms between the state and control in the LQR cost function. The identity matrix in (9.2d) indicates that the coupling matrix between the control and the output is normalized; that is, the control is normalized so that all controls are equally weighted in the cost. This normalization is performed to simplify subsequent derivations, and so that the final results match those in the current literature. Normalization of this coupling matrix can always be accomplished provided  $\mathbf{D}_{yu}$  has full rank.

The  $\mathcal{H}_\infty$  full information control problem is to find a feedback controller, utilizing the state and the disturbance, that minimizes the closed-loop system  $\infty$ -norm:

$$J = \|\mathbf{G}_{yw}\|_{\infty, [0, t_f]} = \sup_{\|w(t)\|_{2, [0, t_f]} \neq 0} \frac{\|y(t)\|_{2, [0, t_f]}}{\|w(t)\|_{2, [0, t_f]}}. \quad (9.3)$$

A block diagram of the full information control system is given in Figure 9.1 where the controller is a linear system (not necessarily time-invariant) denoted  $\mathcal{K}(\bullet)$ . Note that defining the cost in terms of the  $\infty$ -norm implies that the initial state is zero:<sup>4</sup>

$$x(0) = 0.$$

<sup>3</sup>These assumptions can be relaxed. In general, the plant need only be detectable from the reference output and stabilizable from the control input.

<sup>4</sup>In applications, the initial state may not be zero. The  $\mathcal{H}_\infty$  full information controller typically yields a reasonable transient response, but this is not guaranteed and must be verified. Applying weights to the various terms in the reference output can be used to modify the transient response when necessary.



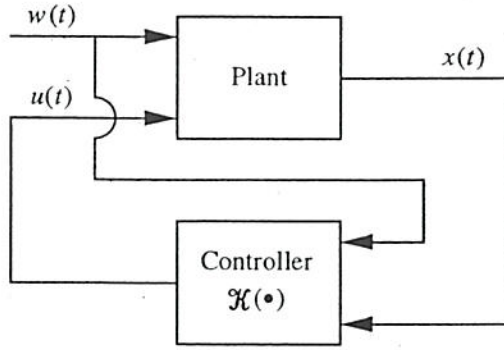


FIGURE 9.1 Full information control

The  $\infty$ -norm can be defined on either a finite or infinite ( $t_f = \infty$ ) time interval. When the time interval is infinite, the controller must also internally stabilize the closed-loop system.

A suitable objective function is required in order to apply differential game theory to the solution of the full information control problem. The  $\infty$ -norm cost function (9.3) is not acceptable as an objective function, since this cost only depends on the controller; that is, the supremum makes this function independent of a particular disturbance input. Dropping the supremum from the cost function (9.3) yields a suitable objective function. While theoretically acceptable, the solution of the differential game with this objective function is intractable, in general.

A quadratic objective function, which yields tractable solutions of the differential game, can be obtained by considering the bound on the closed-loop  $\infty$ -norm:

$$\|G_{yw}\|_{\infty, [0, t_f]} = \sup_{\|w(t)\|_{2, [0, t_f]} \neq 0} \frac{\|y(t)\|_{2, [0, t_f]}}{\|w(t)\|_{2, [0, t_f]}} < \gamma, \quad (9.4)$$

where  $\gamma$  is called the *performance bound*. A controller that satisfies this bound is called a suboptimal solution to the  $\mathcal{H}_\infty$  full information control problem, or simply a suboptimal controller. The suboptimal controller must also satisfy the bound obtained by squaring (9.4):

$$\|G_{yw}\|_{\infty, [0, t_f]}^2 = \sup_{\|w(t)\|_{2, [0, t_f]} \neq 0} \left\{ \frac{\|y(t)\|_{2, [0, t_f]}^2}{\|w(t)\|_{2, [0, t_f]}^2} \right\} < \gamma^2.$$

For the supremum to satisfy this strict inequality, the term within the curly brackets must be bounded away from  $\gamma^2$ ; that is, for some  $\varepsilon$ ,

$$\frac{\|y(t)\|_{2, [0, t_f]}^2}{\|w(t)\|_{2, [0, t_f]}^2} \leq \gamma^2 - \varepsilon^2.$$

Multiplying both sides by the denominator, and grouping the resulting terms yields

$$\|y(t)\|_{2, [0, t_f]}^2 - \gamma^2 \|w(t)\|_{2, [0, t_f]}^2 \leq -\varepsilon^2 \|w(t)\|_{2, [0, t_f]}^2. \quad (9.5)$$

Note that satisfaction of this inequality for all disturbance inputs and some  $\varepsilon$  is equivalent to the bound on the closed-loop  $\infty$ -norm (9.4). The expression on the left of (9.5) can be used as an objective function:

$$J_\gamma(x, u, w) = \|y(t)\|_2^2 - \gamma^2 \|w(t)\|_2^2. \quad (9.6)$$

Differential game theory can then be applied to generate a control that minimizes this objective function in the presence of the worst-case disturbance. If the minimum of the objective function satisfies (9.5), then the bound on the closed-loop  $\infty$ -norm is achieved.

This differential game is posed as an open-loop mini-max problem; that is, the control input is not required to be generated by a feedback controller. Therefore, this problem allows more general control inputs than those allowed in the  $\mathcal{H}_\infty$  suboptimal control problem. As will be seen, the solution of the differential game does yield a full information feedback control, making it a valid candidate solution of the  $\mathcal{H}_\infty$  suboptimal control problem. In fact, the saddle point of the differential game is an  $\mathcal{H}_\infty$  suboptimal full information controller. This fact is demonstrated after first deriving the solution of the differential game, which is referred to as an  $\mathcal{H}_\infty$  suboptimal full information controller in anticipation of demonstrating this equivalence.

General conditions can be given for the existence of  $\mathcal{H}_\infty$  suboptimal controllers. The  $\mathcal{H}_\infty$  optimal controller can then be approximated by decreasing the performance bound until a suboptimal controller no longer exists. As a matter of practice, a binary search with a Cauchy convergence criteria can be used when decreasing this bound, allowing arbitrarily close approximation of the  $\mathcal{H}_\infty$  optimal solution.

### 9.2.1 The Hamiltonian Equations

The differential game specified by the objective function (9.6) and the game dynamics (9.2) yields a constrained mini-max problem. An unconstrained mini-max problem of higher dimension can be obtained by appending the constraint to the objective function:

$$J_{a,\gamma}(u, w, p) = \int_0^{t_f} y^T(t)y(t) - \gamma^2 w^T(t)w(t) \\ + 2p^T(t)\{Ax(t) + B_u u(t) + B_w w(t) - \dot{x}(t)\}dt,$$

where the Lagrange multiplier term  $p(t)$  is referred to as the *costate* (following the terminology presented for the LQR). The factor of 2 multiplying the constraint equation is used to simplify the final results. Note that 2 times a constraint equation is also a constraint equation.

A necessary condition for a saddle point is that the variation of the augmented objective function equal zero. The increment of the augmented cost function is

$$\Delta J_{a,\gamma}(u, w, p, \delta u, \delta w, \delta p) \\ = J_{a,\gamma}(u + \delta u, w + \delta w, p + \delta p) - J_{a,\gamma}(u, w, p) \\ = \int_0^{t_f} (x + \delta x)^T C_y^T C_y (x + \delta x) + (u + \delta u)^T (u + \delta u) - \gamma^2 (w + \delta w)^T (w + \delta w) \\ + 2(p + \delta p)^T \{A(x + \delta x) + B_u (u + \delta u) + B_w (w + \delta w) - (\dot{x} + \delta \dot{x})\} dt \\ - \int_0^{t_f} x^T C_y^T C_y x + u^T u - \gamma^2 w^T w + 2p^T \{Ax + B_u u + B_w w - \dot{x}\} dt,$$



where (9.2b) has been used to substitute for  $y(t)$ , the fact that  $\mathbf{D}_{yu}^T \mathbf{D}_{yu} = \mathbf{I}$  has been used, and the time indexes have been omitted to simplify the notation. Expanding this expression and grouping terms, we have

$$\begin{aligned} \Delta J_{a,\gamma} = & \int_0^{t_f} \delta x^T \mathbf{C}_y^T \mathbf{C}_y \delta x + \delta u^T \delta u - \gamma^2 \delta w^T \delta w \\ & + 2\delta p^T \{ \mathbf{A} \delta x + \mathbf{B}_u \delta u + \mathbf{B}_w \delta w - \delta \dot{x} \} + 2u^T \delta u \\ & + 2x^T \mathbf{C}_y^T \mathbf{C}_y \delta x - 2\gamma^2 w^T \delta w + 2\delta p^T \{ \mathbf{A} x + \mathbf{B}_u u + \mathbf{B}_w w - \dot{x} \} \\ & + 2p^T \{ \mathbf{A} \delta x + \mathbf{B}_u \delta u + \mathbf{B}_w \delta w - \delta \dot{x} \} dt. \end{aligned}$$

A necessary condition for the trajectory  $x(t)$ ,  $p(t)$ ,  $u(t)$ , and  $w(t)$  to be a saddle point is that the variation of the objective function equal zero:

$$\begin{aligned} \delta J_{a,\gamma}(u, w, p, \delta u, \delta w, \delta p) = & \int_0^{t_f} 2x^T \mathbf{C}_y^T \mathbf{C}_y \delta x + 2u^T \delta u - 2\gamma^2 w^T \delta w \\ & + 2\delta p^T \{ \mathbf{A} x + \mathbf{B}_u u + \mathbf{B}_w w - \dot{x} \} \\ & + 2p^T \{ \mathbf{A} \delta x + \mathbf{B}_u \delta u + \mathbf{B}_w \delta w - \delta \dot{x} \} dt = 0. \end{aligned} \quad (9.7)$$

Integration by parts yields

$$\int_0^{t_f} p^T(t) \delta \dot{x}(t) dt = p^T(t_f) \delta x(t_f) - p^T(0) \delta x(0) - \int_0^{t_f} \dot{p}^T(t) \delta x(t) dt, \quad (9.8)$$

which can be used to eliminate the variation of the state derivative in (9.7). Further, the state initial condition is fixed, so  $\delta x(0) = 0$ . Substituting this result and (9.8) into (9.7), and regrouping terms yields the necessary condition for a saddle point:

$$\begin{aligned} \delta J_{a,\gamma}(u, w, p, \delta u, \delta w, \delta p) = & -2p^T(t_f) \delta x(t_f) + \int_0^{t_f} (2\dot{p}^T + 2x^T \mathbf{C}_y^T \mathbf{C}_y + 2p^T \mathbf{A}) \delta x \\ & + (2p^T \mathbf{B}_w - 2\gamma^2 w^T) \delta w + (2u^T + 2p^T \mathbf{B}_u) \delta u \\ & + 2\delta p^T (\mathbf{A} x + \mathbf{B}_u u + \mathbf{B}_w w - \dot{x}) dt = 0. \end{aligned}$$

Since the variations  $\delta x(t_f)$ ,  $\delta x$ ,  $\delta u$ , and  $\delta p$  are all arbitrary, this expression is only equal to zero if

$$p(t_f) = 0; \quad (9.9a)$$

$$\dot{p}(t) = -\mathbf{C}_y^T \mathbf{C}_y x(t) - \mathbf{A}^T p(t); \quad (9.9b)$$

$$u(t) = -\mathbf{B}_u^T p(t); \quad (9.9c)$$

$$w(t) = \gamma^{-2} \mathbf{B}_w^T p(t); \quad (9.9d)$$

$$\dot{x}(t) = \mathbf{A} x(t) + \mathbf{B}_u u(t) + \mathbf{B}_w w(t). \quad (9.9e)$$

Eliminating  $u(t)$  and  $w(t)$  from (9.9b) and (9.9e), and combining the resulting equations into a single state equation yields

$$\begin{bmatrix} \dot{x}(t) \\ \dot{p}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{A} & -\mathbf{B}_u \mathbf{B}_u^T + \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T \\ -\mathbf{C}_y^T \mathbf{C}_y & -\mathbf{A}^T \end{bmatrix} \begin{bmatrix} x(t) \\ p(t) \end{bmatrix} = \mathcal{X}_\infty \begin{bmatrix} x(t) \\ p(t) \end{bmatrix}. \quad (9.10)$$

This is the Hamiltonian system for the full information  $\mathcal{H}_\infty$  control problem, and the matrix  $\mathcal{H}_\infty$  is called the Hamiltonian.

The  $\mathcal{H}_\infty$  suboptimal control can be found by solving the Hamiltonian system subject to the final condition

$$p(t_f) = 0.$$

The solution of the Hamiltonian system (9.10), at the final time, given an initial condition at time  $t$ , is

$$\begin{bmatrix} x(t_f) \\ p(t_f) \end{bmatrix} = e^{\mathcal{H}_\infty(t_f-t)} \begin{bmatrix} x(t) \\ p(t) \end{bmatrix} = \begin{bmatrix} \Phi_{11}(t_f-t) & \Phi_{12}(t_f-t) \\ \Phi_{21}(t_f-t) & \Phi_{22}(t_f-t) \end{bmatrix} \begin{bmatrix} x(t) \\ p(t) \end{bmatrix}.$$

Substituting the final condition (9.9a) into this equation yields

$$\begin{bmatrix} x(t_f) \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi_{11}(t_f-t) & \Phi_{12}(t_f-t) \\ \Phi_{21}(t_f-t) & \Phi_{22}(t_f-t) \end{bmatrix} \begin{bmatrix} x(t) \\ p(t) \end{bmatrix}.$$

The lower block of this matrix equation can be solved for the costate in terms of the following state:

$$p(t) = -\{\Phi_{22}(t_f-t)\}^{-1}\Phi_{21}(t_f-t)x(t) = \mathbf{P}(t)x(t), \quad (9.11)$$

where  $\mathbf{P}(t)$  is the matrix of proportionality between the costate and the state. This matrix of proportionality is fully specified by the state-transition matrix of the Hamiltonian system, provided the inverse in (9.11) exists at all times between the initial time and the final time.

The  $\mathcal{H}_\infty$  suboptimal control is then found from (9.9c):

$$u(t) = -\mathbf{B}_u^T \mathbf{P}(t)x(t) = \mathbf{B}_u^T \{\Phi_{22}(t_f-t)\}^{-1} \Phi_{21}(t_f-t)x(t) = -\mathbf{K}(t)x(t), \quad (9.12)$$

where  $\mathbf{K}(t)$  is termed the  $\mathcal{H}_\infty$  suboptimal feedback gain matrix. While this control is generated from the necessary conditions for the solution of the differential game, it in fact specifies a solution of the differential game (provided the inverse exists at all times). Further, this control is given by a linear, time-varying, state feedback law, making it a candidate solution of the  $\mathcal{H}_\infty$  full information suboptimal control problem. Indeed, this feedback law is a solution of this suboptimal control problem, which will be demonstrated after first presenting an example, and showing that the matrix  $\mathbf{P}(t)$  can be found as the solution of a Riccati equation.

**EXAMPLE 9.1** Let the plant be modeled as follows:

$$\begin{aligned} \dot{x}(t) &= x(t) + u(t) + 2w(t); \\ y(t) &= \begin{bmatrix} 10 \\ 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t), \end{aligned}$$

and the objective function be

$$J_\gamma\{x(t), u(t), w(t)\} = \|y(t)\|_2^2 - \gamma^2 \|w(t)\|_2^2.$$



Note that state and control weightings are incorporated into the output equation, which is scaled to yield a unity weight on the control. The Hamiltonian system is

$$\begin{bmatrix} \dot{x}(t) \\ \dot{p}(t) \end{bmatrix} = \begin{bmatrix} 1 & -1 + 4\gamma^{-2} \\ -100 & -1 \end{bmatrix} \begin{bmatrix} x(t) \\ p(t) \end{bmatrix},$$

which has the state-transition matrix

$$\begin{aligned} \Phi(t) &= e^{\mathcal{Z}_\infty t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathcal{Z}_\infty)^{-1}\} \\ &= \mathcal{L}^{-1}\left\{\frac{1}{s^2 - 101 + 400\gamma^{-2}} \begin{bmatrix} s + 1 & -1 + 4\gamma^{-2} \\ -100 & s - 1 \end{bmatrix}\right\}. \end{aligned}$$

A saddle point of this differential game exists (and an  $\mathcal{H}_\infty$  full information suboptimal controller exists) for a given performance bound provided  $\Phi_{22}(t_f - t)$  has an inverse throughout the time interval; that is, the scalar  $\Phi_{22}(t_f - t) \neq 0$ . This element of the state transition matrix can be computed:

$$\Phi_{22}(t) = \mathcal{L}^{-1}\left\{\frac{s - 1}{s^2 - 101 + 400\gamma^{-2}}\right\} = \begin{cases} \frac{a - 1}{2a}e^{at} + \frac{a + 1}{2a}e^{-at} & \gamma > \sqrt{\frac{400}{101}} \\ 1 - t & \gamma = \sqrt{\frac{400}{101}} \\ \frac{\sqrt{\omega^2 + 1}}{\omega} \sin(\omega t + \theta) & \gamma < \sqrt{\frac{400}{101}} \end{cases}$$

where  $a = \sqrt{101 - 400\gamma^{-2}}$ ,  $\omega = \sqrt{-101 + 400\gamma^{-2}}$ , and  $\theta = -\tan^{-1}(\omega)$ . Note that as  $\gamma$  gets smaller,  $a$  gets smaller,  $\omega$  gets bigger, and  $\theta$  gets more negative. When  $\gamma < \sqrt{400/101}$ , the eigenvalues of the Hamiltonian matrix are imaginary, and the gain does not exist for long time intervals, since  $\Phi_{22}(t)$  periodically crosses zero. Further, the fact that the feedback gain does not exist implies that there is no solution of the differential game (and no solution of the  $\mathcal{H}_\infty$  suboptimal control problem) for the given bound. Less obvious from these expressions is that the time interval over which a feedback gain exists becomes monotonically larger as  $\gamma$  is increased, reaching infinity for  $\gamma > 2$ .

For the bound  $\gamma = 2.03$  ( $a = 2$ ), and the final time  $t_f = 3$ , the  $\mathcal{H}_\infty$  suboptimal feedback gain is

$$K(t) = \frac{100e^{2(3-t)} - 100e^{-2(3-t)}}{e^{2(3-t)} + 3e^{-2(3-t)}},$$

which is plotted in Figure 9.2. Note that the gain exhibits a transient and then approaches a steady-state value far from the final time. In situations where the time interval is long compared to the settling time of this transient, it may be reasonable to use only the steady-state gain.  $\blacklozenge$

An  $\mathcal{H}_\infty$  suboptimal controller fails to exist over long time intervals, in the above example, when the Hamiltonian has purely imaginary eigenvalues. This result is a general property of  $\mathcal{H}_\infty$  suboptimal control:

The feedback gain exists for arbitrary time intervals, and there is a solution to the  $\mathcal{H}_\infty$  suboptimal control problem, only if the Hamiltonian matrix has no purely imaginary eigenvalues.

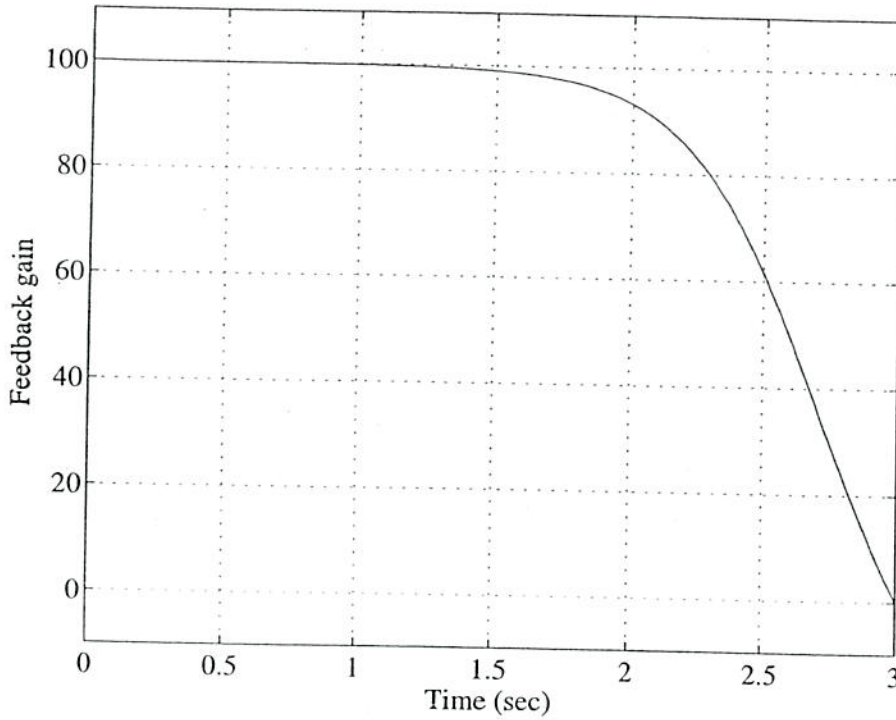


FIGURE 9.2 The feedback gain for Example 9.1

Note that this is an “only if” statement. The fact that the Hamiltonian matrix has no purely imaginary eigenvalues is not sufficient to guarantee existence of a solution to the steady-state  $\mathcal{H}_\infty$  suboptimal control problem. Additional conditions for the existence of this steady-state control are presented in subsection 9.2.4.

### 9.2.2 The Riccati Equation

The solution of the Hamiltonian system for the linear quadratic regulator was shown in Chapter 6 to be related to a nonlinear matrix differential equation, known as a Riccati equation. The Hamiltonian system for the  $\mathcal{H}_\infty$  suboptimal control problem can also be related to a Riccati equation. This Riccati equation has only final conditions and can be solved backward in time using any numerical integration package.

The solution of the  $\mathcal{H}_\infty$  suboptimal control problem can be reduced to finding the matrix  $\mathbf{P}(t)$ , since the feedback gain (9.12) only depends on this matrix and  $\mathbf{B}_u$ . A differential equation for  $\mathbf{P}(t)$  can be generated by taking the derivative of (9.11):

$$\dot{p}(t) = \dot{\mathbf{P}}(t)x(t) + \mathbf{P}(t)\dot{x}(t).$$

Substituting for  $\dot{p}(t)$  and  $\dot{x}(t)$ , using the Hamiltonian system (9.10) yields

$$-\mathbf{C}_y^T \mathbf{C}_y x(t) - \mathbf{A}^T p(t) = \dot{\mathbf{P}}(t)x(t) + \mathbf{P}(t)\{\mathbf{A}x(t) - (\mathbf{B}_u \mathbf{B}_u^T - \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T)p(t)\}.$$

Then, substituting for  $p(t)$  using (9.11) and rearranging yields

$$\{\dot{\mathbf{P}}(t) + \mathbf{P}(t)\mathbf{A} + \mathbf{A}^T \mathbf{P}(t) + \mathbf{C}_y^T \mathbf{C}_y - \mathbf{P}(t)(\mathbf{B}_u \mathbf{B}_u^T - \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T)\mathbf{P}(t)\}x(t) = 0.$$



This equation is valid for any state  $x(t)$ ; therefore, the matrix in curly brackets must equal zero:

$$\dot{\mathbf{P}}(t) = -\mathbf{P}(t)\mathbf{A} - \mathbf{A}^T\mathbf{P}(t) - \mathbf{C}_y^T\mathbf{C}_y + \mathbf{P}(t)(\mathbf{B}_u\mathbf{B}_u^T - \gamma^{-2}\mathbf{B}_w\mathbf{B}_w^T)\mathbf{P}(t). \quad (9.13)$$

This equation is the Riccati differential equation for the  $\mathcal{H}_\infty$  suboptimal control problem.

The Riccati solution  $\mathbf{P}(t)$  is a symmetric matrix (if it exists), which can be found by solving (9.13) backward in time from the final condition:

$$\mathbf{P}(t_f) = \mathbf{0}. \quad (9.14)$$

This final condition is obtained by letting  $p(t_f) = 0$  in (9.11) and recognizing that the result is valid for any final state. The fact that the Riccati solution is symmetric is shown by noting that the derivative is symmetric whenever the Riccati solution is symmetric:

$$\begin{aligned} \dot{\mathbf{P}}^T(t) &= \{-\mathbf{P}(t)\mathbf{A} - \mathbf{A}^T\mathbf{P}(t) - \mathbf{C}_y^T\mathbf{C}_y + \mathbf{P}(t)(\mathbf{B}_u\mathbf{B}_u^T - \gamma^{-2}\mathbf{B}_w\mathbf{B}_w^T)\mathbf{P}(t)\}^T \\ &= -\mathbf{A}^T\mathbf{P}(t) - \mathbf{P}(t)\mathbf{A} - \mathbf{C}_y^T\mathbf{C}_y + \mathbf{P}(t)(\mathbf{B}_u\mathbf{B}_u^T - \gamma^{-2}\mathbf{B}_w\mathbf{B}_w^T)\mathbf{P}(t) = \dot{\mathbf{P}}(t). \end{aligned}$$

Therefore, integrating this derivative backward in time from the final condition yields a symmetric Riccati solution.

### **XAMPLE 9.1 CONTINUED**

The Riccati equation associated with the differential game (and  $\mathcal{H}_\infty$  suboptimal control problem) in Example 9.1 is:

$$\dot{P}(t) = -2P(t) + (1 - 4\gamma^{-2})P^2(t) - 100.$$

The final condition for this differential equation is  $P(3) = 0$ . The solution to this final value problem is

$$P(t) = K(t) = \frac{100e^{2(3-t)} - 100e^{-2(3-t)}}{e^{2(3-t)} + 3e^{-(3-t)}}.$$

This solution can be verified by substitution into the Riccati equation and by evaluating the solution at the final time. Note that the feedback gain equals the Riccati solution, in this example, since  $\mathbf{B}_u = 1$ .  $\blacklozenge$

### **9.2.3 The Value of the Objective Function**

A feedback controller and a worst-case disturbance result when applying Lagrange multipliers and variation theory to the differential game with dynamics (9.2) and objective function (9.6). The feedback gain is found from the solution of either a Hamiltonian or a Riccati equation. All of these results are based on the variation of the objective function, which only specifies necessary conditions for the existence of a saddle point. In fact, the given control and worst-case disturbance form a saddle point of the differential game, and this control is an  $\mathcal{H}_\infty$  suboptimal controller. These two results can be verified by cleverly rewriting the objective function.

To begin, note that

$$\int_0^{t_f} \frac{dx^T(t)\mathbf{P}(t)x(t)}{dt} dt = x^T(t_f)\mathbf{P}(t_f)x(t_f) - x^T(0)\mathbf{P}(0)x(0) = 0,$$

since  $\mathbf{P}(t_f) = 0$  and  $x(0) = 0$ . This integral can be added to the objective function (to complete the square) without changing the value of this function:

$$\begin{aligned} J_\gamma &= \int_0^{t_f} x^T(t)\mathbf{C}_y^T\mathbf{C}_y x(t) + u^T(t)u(t) - \gamma^2 w^T(t)w(t) + \frac{dx^T(t)\mathbf{P}(t)x(t)}{dt} dt \\ &= \int_0^{t_f} x^T(t)\mathbf{C}_y^T\mathbf{C}_y x(t) + u^T(t)u(t) - \gamma^2 w^T(t)w(t) \\ &\quad + \dot{x}^T(t)\mathbf{P}x(t) + x^T(t)\dot{\mathbf{P}}(t)x(t) + x^T(t)\mathbf{P}(t)\dot{x}(t) dt. \end{aligned}$$

Substituting for the derivative of the state using (9.2a) and regrouping terms yields

$$\begin{aligned} J_\gamma &= \int_0^{t_f} x^T(\dot{\mathbf{P}} + \mathbf{C}_y^T\mathbf{C}_y + \mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A})x + u^T u - \gamma^2 w^T w + (\mathbf{B}_u u + \mathbf{B}_w w)^T \mathbf{P}x \\ &\quad + x^T \mathbf{P}(\mathbf{B}_u u + \mathbf{B}_w w) dt, \end{aligned}$$

where the time argument has been dropped to simplify the notation.

Substituting for  $(\dot{\mathbf{P}} + \mathbf{C}_y^T\mathbf{C}_y + \mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A})$  using the Riccati equation (9.13) gives

$$\begin{aligned} J_\gamma &= \int_0^{t_f} x^T \mathbf{P}(\mathbf{B}_u \mathbf{B}_u^T - \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T) \mathbf{P}x + u^T u - \gamma^2 w^T w \\ &\quad + (\mathbf{B}_u u + \mathbf{B}_w w)^T \mathbf{P}x + x^T \mathbf{P}(\mathbf{B}_u u + \mathbf{B}_w w) dt. \end{aligned}$$

Regrouping again yields

$$\begin{aligned} J_\gamma &= \int_0^{t_f} (u + \mathbf{B}_u^T \mathbf{P}x)^T (u + \mathbf{B}_u^T \mathbf{P}x) - \gamma^2 (w - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}x)^T (w - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}x) dt \\ &= \int_0^{t_f} \|u + \mathbf{B}_u^T \mathbf{P}x\|_E^2 - \gamma^2 \|w - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}x\|_E^2 dt \\ &= \|u + \mathbf{B}_u^T \mathbf{P}x\|_{2,[0,t_f]}^2 - \gamma^2 \|w - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}x\|_{2,[0,t_f]}^2, \end{aligned} \quad (9.15)$$

where the norms in the final expression are signal norms. The objective function has a value of zero when the control and disturbance inputs are given by the necessary conditions:

$$u(t) = -\mathbf{B}_u^T \mathbf{P}(t)x(t); \quad (9.16)$$

$$w(t) = \gamma^{-2} \mathbf{B}_w^T \mathbf{P}(t)x(t). \quad (9.17)$$

This point is a saddle point since any other disturbance input, with control (9.16), yields a decrease in the objective value, and any other control input, with disturbance (9.17) yields an increase in the objective value.

To verify that the controller (9.16) is an  $\mathcal{H}_\infty$  suboptimal controller, it must be shown (9.5) that

$$\begin{aligned} J_\gamma &= \|u(t) + \mathbf{B}_u^T \mathbf{P}x(t)\|_{2,[0,t_f]}^2 - \gamma^2 \|w(t) - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}x(t)\|_{2,[0,t_f]}^2 \\ &\leq -\varepsilon^2 \|w(t)\|_{2,[0,t_f]}^2 \end{aligned} \quad (9.18)$$



for some positive  $\varepsilon$ . When using the controller (9.16), (9.18) reduces to

$$J_\gamma = -\gamma^2 \|w(t) - \gamma^{-2} \mathbf{B}_w^T \mathbf{P} x(t)\|_{2,[0,t_f]}^2 \leq -\varepsilon^2 \|w(t)\|_{2,[0,t_f]}^2,$$

or, equivalently,

$$\|w(t)\|_{2,[0,t_f]}^2 \leq \frac{\gamma^2}{\varepsilon^2} \|w(t) - \gamma^{-2} \mathbf{B}_w^T \mathbf{P} x(t)\|_{2,[0,t_f]}^2. \quad (9.19)$$

The fact that a positive  $\varepsilon$  exists that satisfies this equation can be developed by noting that the disturbance input can be generated by a time-varying state model from the input  $[w(t) - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}(t)x(t)]$ :<sup>5</sup>

$$\dot{x}(t) = [\mathbf{A} - \mathbf{B}_u \mathbf{B}_u^T \mathbf{P}(t) + \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T \mathbf{P}(t)]x(t) + \mathbf{B}_w [w(t) - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}(t)x(t)]; \quad (9.20a)$$

$$w(t) = -\gamma^{-2} \mathbf{B}_w^T \mathbf{P}(t)x(t) + [w(t) - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}(t)x(t)]. \quad (9.20b)$$

Since all matrices in this state model are finite, the output is bounded; see Appendix equation (A9.6):

$$\|w(t)\|_{2,[0,t_f]}^2 \leq \|\mathbf{G}_{w\Delta w}\|_{\infty,[0,t_f]}^2 \|w(t) - \gamma^{-2} \mathbf{B}_w^T \mathbf{P}(t)x(t)\|_{2,[0,t_f]}^2,$$

where  $\mathbf{G}_{w\Delta w}$  denotes the state model (9.20). Comparing this result with (9.19), a positive  $\varepsilon$  that satisfies (9.19) is  $\varepsilon = \gamma / \|\mathbf{G}_{w\Delta w}\|_{\infty,[0,t_f]}$  and the control law (9.16) is therefore a solution of the suboptimal  $\mathcal{H}_\infty$  control problem.

The suboptimal control exists provided the Riccati equation (9.13) has a solution over the entire time interval from 0 to  $t_f$ . This control depends on the performance bound selected. Note that the Riccati equation (9.13) reduces to the LQR Riccati equation (6.20) when the performance bound approaches infinity. The LQR Riccati equation is guaranteed to have a solution, indicating that the  $\mathcal{H}_\infty$  Riccati equation has a solution for sufficiently large performance bounds.

The control law (9.16) has been shown to satisfy the infinity norm bound (9.6), and to exist whenever the Riccati equation (9.13) has a solution. The following question immediately comes to mind: If this Riccati equation has no solution, is it possible to find a full information controller that satisfies the bound (9.6)? The answer to this question is no! The existence of a solution to the Riccati equation (9.13) is both necessary and sufficient for the existence of a solution to the  $\mathcal{H}_\infty$  suboptimal control problem. Therefore, the  $\mathcal{H}_\infty$  optimal solution can be approximated to an arbitrary degree of closeness by decreasing the bound until a Riccati solution no longer exists. A formal derivation of the fact that the Riccati solution is required for the existence of a suboptimal full information control can be found in [1], page 224.

The  $\mathcal{H}_\infty$  suboptimal controller presented is not unique. A family of controllers can be generated that all satisfy the bound (9.6), whenever a solution exists. The controller

<sup>5</sup>This model can be obtained by inverting the state model for the system that generates  $(w - \gamma^{-2} \mathbf{B}_w^T \mathbf{P} x)$  from  $w$  (see Exercise 9.7).

presented above is often referred to as the central, or minimum entropy, controller. The generation of this family of controllers is discussed in [1], page 232, and [2], page 431.

#### 9.2.4 Steady-State Full Information Control

The feedback gain in Example 9.1 approaches a steady-state value far from the final time. When operating over infinite time intervals, all finite times are infinitely far from the final time. The time-invariant controller that utilizes the steady-state gain is therefore a solution to the  $\mathcal{H}_\infty$  suboptimal control problem when the time interval is infinite (provided this gain internally stabilizes the system). This time-invariant, full information control can also be utilized to simplify controller implementation when operating over finite time intervals. In this case, the steady-state controller may not satisfy the  $\infty$ -norm bound, but is probably reasonably close provided the time interval is long compared to the gain settling time. These observations, made concerning Example 9.1, can be applied to general  $\mathcal{H}_\infty$  suboptimal control problems.

The existence of a steady-state solution to the  $\mathcal{H}_\infty$  suboptimal control problem is not guaranteed for all values of the  $\infty$ -norm bound (as seen in Example 9.1). A solution does exist for  $\infty$ -norm bounds sufficiently large, given that the plant is controllable from the control input and observable from the reference output (conditions included in the  $\mathcal{H}_\infty$  suboptimal control problem specification). This fact can be deduced by noting that the  $\mathcal{H}_\infty$  Riccati equation approaches the LQR Riccati equation as the performance bound approaches infinity. The above observability and controllability conditions are sufficient to guarantee the existence of a steady-state solution to the LQR and therefore also to guarantee the existence of a steady-state  $\mathcal{H}_\infty$  suboptimal control for sufficiently large performance bounds.

For smaller bounds, the question immediately arises: Under what conditions does a steady-state solution to the  $\mathcal{H}_\infty$  suboptimal control problem exist? The answer follows:

A suboptimal solution that internally stabilizes the closed-loop system and bounds the closed-loop  $\infty$ -norm (9.4) exists if and only if there exists a positive semidefinite solution of the algebraic Riccati equation,

$$\mathbf{P}\mathbf{A} + \mathbf{A}^T\mathbf{P} - \mathbf{P}(\mathbf{B}_u\mathbf{B}_u^T - \gamma^{-2}\mathbf{B}_w\mathbf{B}_w^T)\mathbf{P} + \mathbf{C}_y^T\mathbf{C}_y = \mathbf{0}, \quad (9.21)$$

such that

$$\mathbf{A} - (\mathbf{B}_u\mathbf{B}_u^T - \gamma^{-2}\mathbf{B}_w\mathbf{B}_w^T)\mathbf{P} \quad (9.22)$$

is stable, that is, all of the eigenvalues of this matrix have negative real parts. The suboptimal controller is then given as

$$u(t) = -\mathbf{B}_u^T\mathbf{P}x(t) = -\mathbf{K}x(t). \quad (9.23)$$

Note that condition (9.22) is included to guarantee the internal stability of the feedback system.

Equation (9.15) was used to show that the infinity norm is bounded as in (9.4), in the finite time case. The derivation of (9.15) requires the existence of the Riccati



solution at all times during the interval of operation. While the above conditions are sufficient to guarantee the existence of the Riccati solution at all times, this fact is certainly not obvious. Simply taking the limit of (9.15) as the final time approaches infinity is therefore not sufficient to verify that the performance bound is achieved. An alternative method of demonstrating that (9.23) represents a solution of the  $\mathcal{H}_\infty$  suboptimal control problem is presented below.

The fact that (9.23) is a solution of the  $\mathcal{H}_\infty$  suboptimal control problem can be verified by forming the closed-loop state model, verifying that the closed-loop system is stable, and applying the bounded real lemma to verify that the gain satisfies the performance bound. The closed-loop state model is

$$\begin{aligned}\dot{x}(t) &= (\mathbf{A} - \mathbf{B}_u \mathbf{B}_u^T \mathbf{P})x(t) + \mathbf{B}_w w(t) = \mathbf{A}_{cl}x(t) + \mathbf{B}_{cl}w(t); \\ y(t) &= \begin{bmatrix} \mathbf{C}_y \\ -\mathbf{D} \mathbf{B}_u^T \mathbf{P} \end{bmatrix} x(t) + \mathbf{0}w(t) = \mathbf{C}_{cl}x(t) + \mathbf{D}_{cl}w(t).\end{aligned}$$

To show that this system is internally stable, add and subtract  $(\mathbf{P} \mathbf{B}_u \mathbf{B}_u^T \mathbf{P})$  to the algebraic Riccati equation (9.21) and rearrange to give

$$\begin{aligned}\mathbf{P}(\mathbf{A} - \mathbf{B}_u \mathbf{B}_u^T \mathbf{P}) + (\mathbf{A} - \mathbf{B}_u \mathbf{B}_u^T \mathbf{P})^T \mathbf{P} \\ + \gamma^{-2} \mathbf{P} \mathbf{B}_w \mathbf{B}_w^T \mathbf{P} + \mathbf{P} \mathbf{B}_u \mathbf{B}_u^T \mathbf{P} + \mathbf{C}_y^T \mathbf{C}_y = \mathbf{0}.\end{aligned}\tag{9.24}$$

Note that this is a Lyapunov equation:

$$\mathbf{P} \mathbf{A}_{cl} + \mathbf{A}_{cl}^T \mathbf{P} + [\gamma^{-1} \mathbf{P} \mathbf{B}_w \mid \mathbf{P} \mathbf{B}_u \mid \mathbf{C}_y^T] \begin{bmatrix} \gamma^{-1} \mathbf{B}_w^T \mathbf{P} \\ \mathbf{B}_u^T \mathbf{P} \\ \mathbf{C}_y \end{bmatrix} = \mathbf{0}.$$

The existence of a positive semidefinite solution  $\mathbf{P}$  for this equation implies that

$$\lim_{t \rightarrow \infty} \begin{bmatrix} \gamma^{-1} \mathbf{B}_w^T \mathbf{P} \\ \mathbf{B}_u^T \mathbf{P} \\ \mathbf{C}_y \end{bmatrix} e^{\mathbf{A}_{cl} t} = \mathbf{0},\tag{9.25}$$

by Appendix equation (A7.7).

Since the plant is observable from  $y(t)$  (the output associated with  $\mathbf{C}_y$ ), and the controller contains no states, the closed-loop system is observable from  $y(t)$ . Therefore, the closed-loop system is internally stable, since any unstable modes, modes that do not decay to zero, will appear in the output. A formal proof that (9.25) and observability imply stability is given in the Appendix; see Appendix equation (A7.8).

The bounded real lemma is used to demonstrate that the closed-loop system  $\infty$ -norm satisfies the bound (9.4). This lemma states,

The infinity norm of the generic stable system (2.4) is bounded,

$$\|\mathbf{G}(s)\|_\infty < \gamma,\tag{9.26}$$

if and only if

$$\bar{\sigma}(\mathbf{D}) < \gamma,\tag{9.27a}$$

and there exists a symmetric matrix  $\mathbf{P}$  that satisfies the algebraic Riccati equation

$$\begin{aligned} & \mathbf{P}(\mathbf{A} + \mathbf{B}\mathbf{R}^{-1}\mathbf{D}^T\mathbf{C}) + (\mathbf{A} + \mathbf{B}\mathbf{R}^{-1}\mathbf{D}^T\mathbf{C})^T\mathbf{P} \\ & + \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} + \mathbf{C}^T(\mathbf{I} + \mathbf{D}\mathbf{R}^{-1}\mathbf{D}^T)\mathbf{C} = \mathbf{0}, \end{aligned} \quad (9.27b)$$

such that

$$\mathbf{A} + \mathbf{B}\mathbf{R}^{-1}\mathbf{D}^T\mathbf{C} + \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} \quad (9.27c)$$

is stable, that is, has all eigenvalues with negative real parts, where  $\mathbf{R} = \gamma^2\mathbf{I} + \mathbf{D}^T\mathbf{D}$ .

A proof of this lemma can be found in Section A10 of the Appendix. To apply this result, it must be shown that the closed-loop state model satisfies the conditions given in (9.27).

Condition (9.27a) requires that the input-to-output coupling matrix be bounded:

$$\bar{\sigma}(\mathbf{D}_{cl}) = \bar{\sigma}(\mathbf{0}) = 0 < \gamma.$$

Further, conditions (9.21) and (9.22) are equivalent to conditions (9.27b) and (9.27c) when applied to the closed-loop system. To see this, substitute the matrices from the closed-loop state model into (9.27b):

$$\begin{aligned} & \mathbf{P}(\mathbf{A} - \mathbf{B}_u\mathbf{B}_u^T\mathbf{P}) + (\mathbf{A} - \mathbf{B}_u\mathbf{B}_u^T\mathbf{P})^T\mathbf{P} \\ & + \gamma^{-2}\mathbf{P}\mathbf{B}_w\mathbf{B}_w^T\mathbf{P} + [\mathbf{C}^T \quad -\mathbf{P}\mathbf{B}_u\mathbf{D}^T] \begin{bmatrix} \mathbf{C} \\ -\mathbf{D}\mathbf{B}_u^T\mathbf{P} \end{bmatrix} = \mathbf{0}. \end{aligned}$$

Expanding the last term in this equation and remembering that  $\mathbf{D}^T\mathbf{D} = \mathbf{I}$  yields (9.24), which is equivalent to (9.21), as demonstrated above. For the closed-loop system, the matrix in (9.27c) reduces to (9.22):

$$\mathbf{A}_{cl} + \mathbf{B}_{cl}\mathbf{R}^{-1}\mathbf{D}_{cl}^T\mathbf{C}_{cl} + \mathbf{B}_{cl}\mathbf{R}^{-1}\mathbf{B}_{cl}^T\mathbf{P} = \mathbf{A} - \mathbf{B}_u\mathbf{B}_u^T\mathbf{P} + \gamma^{-2}\mathbf{B}_w\mathbf{B}_w^T\mathbf{P},$$

which is required to be stable. Summarizing, for the closed-loop system, (9.27a) is satisfied, and there exists a solution to (9.27b) such that (9.27c) is stable. Together these results imply that the closed-loop system  $\infty$ -norm is bounded as given in (9.26) and equivalently in (9.4). Since the closed-loop system is stable, this implies that the control (9.23) is an  $\mathcal{H}_\infty$  suboptimal controller.

The above presentation does not demonstrate that the conditions relating to (9.21) and (9.22) are necessary for the existence of an  $\mathcal{H}_\infty$  suboptimal controller.<sup>6</sup> But in fact, these conditions are both necessary and sufficient for the existence of such a controller. A formal proof of the necessity of these conditions can be found in [1], page 242.

## EXAMPLE 9.2

An antenna is required to remain pointed at a satellite in the presence of disturbance torques caused by gravity, wind, squirrels climbing on the dish, and so on. A state model for this antenna is

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w(t); \\ e(t) &= [1 \quad 0]x(t), \end{aligned}$$

<sup>6</sup> Actually, given the result in Section A10 of the Appendix, necessity is demonstrated for the state feedback control, but not general full information control.



where  $u(t)$  is the control torque,  $w(t)$  is the disturbance torque, and  $e(t)$  is the tracking error of the antenna. The tracking error of less than 0.1 rad is desired, even in the presence of disturbance torques of up to 2 N-m. The control magnitude is also required to be less than 10 N-m. These specifications can be appended to the plant as weighting functions to yield the model in standard form:

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 10 \end{bmatrix} u_1(t) + \begin{bmatrix} 0 \\ 2 \end{bmatrix} w_1(t); \\ y(t) &= \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_1(t),\end{aligned}$$

where  $w_1(t)$  and  $u_1(t)$  are the normalized disturbance and control inputs, respectively. In this form,  $w_1(t)$  is less than 1 and both elements of  $y(t)$  are required to be less than 1. The original control input can be recovered from the normalized control:

$$u(t) = 10u_1(t).$$

A full information controller is generated for  $\gamma = 1$ :

$$u_1(t) = -[10.2 \quad 1.36]x(t) \Rightarrow u(t) = -[102 \quad 13.6]x(t).$$

The magnitude of the closed-loop system frequency response with this control is given in Figure 9.3. This frequency response is for the unnormalized system. Note that the maximum gain from the disturbance input to the error is 0.01, which is less than the gain of 0.05 required to meet the tracking error specification (provided the input is sinusoidal). In addition, the maximum gain from the disturbance input to the control is 1.26, which is less than the gain of 5 required to maintain a control less than 10 (again, for sinusoidal disturbance inputs). Based on these results, it is expected that the tracking error and control magnitude will meet the specified bounds even in the presence of reasonable nonsinusoidal disturbance inputs, but this is not guaranteed.

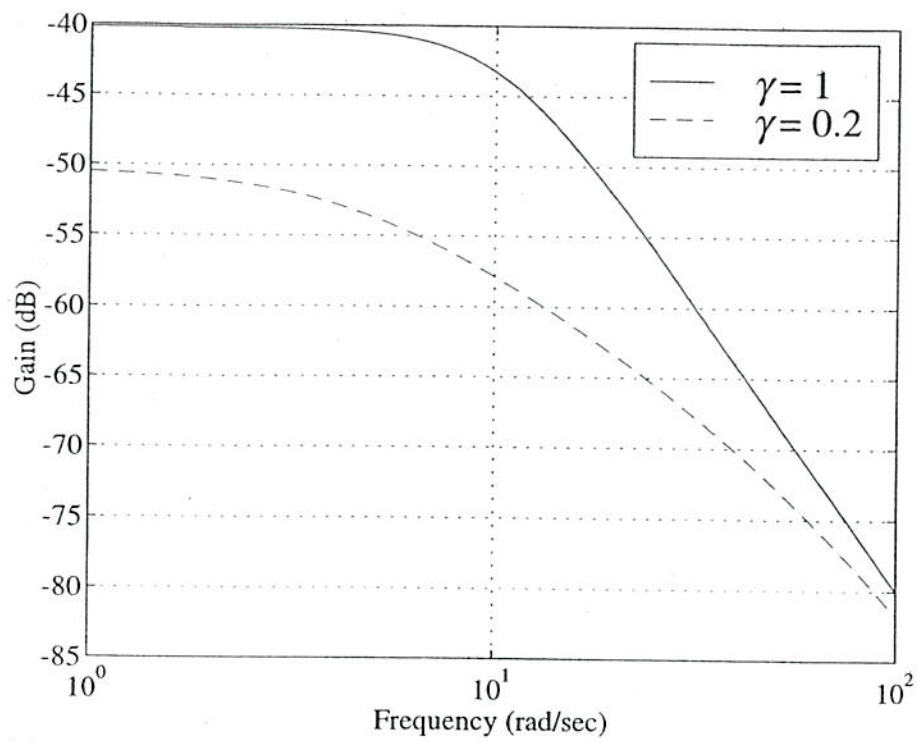
The given controller is designed solely to achieve the specified bounds on the closed-loop system gains. No specifications were included on the transient response of the control system.<sup>7</sup> Therefore, the transient response should be checked to determine if it is acceptable. The closed-loop poles are  $-7.3 \pm 7.0j$ , which should yield a reasonably good transient response.

Smaller values of the performance bound (with  $\gamma > 0.2$ ) can be used to generate additional controllers that meet the specifications. This limit on the performance bound can be obtained by decreasing the bound, solving (9.21), and checking to see if this solution is positive definite and (9.22) is stable. Alternatively, this bound can be generated by noting that the control and the disturbance enter the system at the same point. A full information controller,

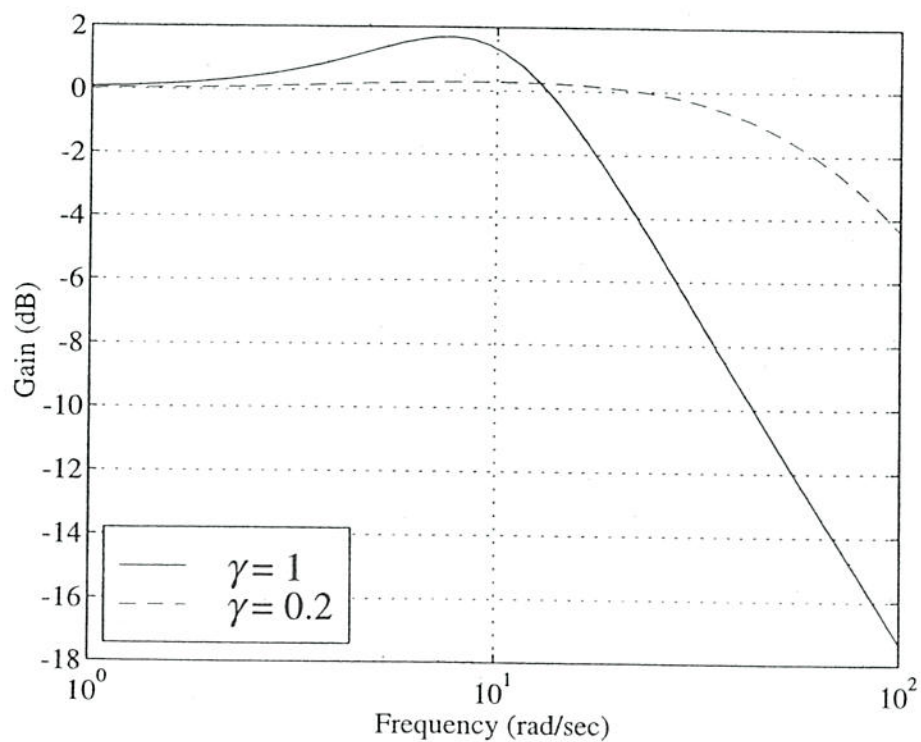
$$u(t) = -w(t),$$

would then yield perfect tracking and satisfy the bound on the control input only for values of  $\gamma > 0.2$ .

<sup>7</sup>Transient response can be mandated, to a degree, in an  $\mathcal{H}_\infty$  setting by specifying the desired closed-loop frequency response.



(a)



(b)

**FIGURE 9.3** Frequency responses for Example 9.2: (a) tracking error; (b) control



Another  $\mathcal{H}_\infty$  suboptimal controller is generated for  $\gamma = 0.21$ :

$$u_1(t) = -[32.8 \quad 7.39]x(t) \Rightarrow u(t) = -[328 \quad 73.9]x(t).$$

The closed-loop frequency response when using this controller is also shown in Figure 9.3. Comparing the two controllers, the use of the smaller performance bound increases the feedback gains, significantly decreases the maximum tracking error, and increases the bandwidth of the transfer function from the disturbance input to the control. This increase in bandwidth is also reflected in the closed-loop poles  $\{-4.7, -70\}$ . These poles are indicative of an overdamped, but probably acceptable, transient response.

The weights on the control and tracking errors can be adjusted individually to trade off tracking accuracy and control magnitude. The results of these changes are analogous to the results obtained when changing the weighting matrices in the LQR. ♦

The bandwidth of the gain between the disturbance input and the control input was observed to increase when approaching the  $\mathcal{H}_\infty$  optimal solution in this example. This tends to be a general property of  $\mathcal{H}_\infty$  control. This increase in bandwidth is often detrimental to the overall control system, since many actuators have finite bandwidths, and controller dynamics (which may have been ignored) become more significant at higher frequencies. Therefore, it is often desirable to use an  $\mathcal{H}_\infty$  suboptimal controller that just meets (or exceeds by a safe margin of error) the specifications rather than attempt to optimize the infinity norm.

**Computation of the Steady-State  $\mathcal{H}_\infty$  Full Information Control** The steady-state solution of the algebraic Riccati equation can be found from the eigensystem decomposition of the Hamiltonian matrix:

$$\mathbf{P} = \Psi_{21}(\Psi_{11})^{-1},$$

where

$$\begin{bmatrix} \Psi_{11} \\ \hline \Psi_{21} \end{bmatrix}$$

is a matrix whose columns are the eigenvectors of the Hamiltonian associated with the stable eigenvalues. The derivation of this result is analogous to that presented for the LQR and is therefore omitted. The steady-state Riccati solution is then used to generate the steady-state full information controller as given in (9.23).

**Existence Results in Terms of the Hamiltonian** The existence of an  $\mathcal{H}_\infty$  suboptimal control can also be specified in terms of the Hamiltonian matrix:

A suboptimal solution that stabilizes the system and bounds the  $\infty$ -norm of the closed-loop transfer function (9.4) exists if and only if the Hamiltonian matrix,

$$\mathcal{H}_\infty = \begin{bmatrix} \mathbf{A} & \vdots & -\mathbf{B}_u \mathbf{B}_u^T + \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T \\ \hline -\mathbf{C}_y^T \mathbf{C}_y & \vdots & -\mathbf{A}^T \end{bmatrix}, \quad (9.28)$$

has no eigenvalues on the imaginary axis,  $\Psi_{11}$  is invertible, and

$$\mathbf{P} = \Psi_{21}(\Psi_{11})^{-1} \geq \mathbf{0}. \quad (9.29)$$

The suboptimal controller is then given as

$$u(t) = -\mathbf{B}_u^T \Psi_{21}(\Psi_{11})^{-1} x(t) = -\mathbf{K}x(t). \quad (9.30)$$

The fact that no  $\mathcal{H}_\infty$  suboptimal control exists when the Hamiltonian has eigenvalues on the imaginary axis can be intuitively understood by considering (9.12). Purely imaginary eigenvalues give rise to undamped oscillations in the state-transition matrix of the Hamiltonian. These oscillations, in turn, yield oscillations in the time-varying Riccati solution, thus guaranteeing that a limit does not exist.

The above result is simply a restatement of the previous existence result for the  $\mathcal{H}_\infty$  suboptimal control problem based on the relationship between the Hamiltonian and the steady-state Riccati solution. This formulation of the existence result is presented for completeness and to aid in understanding related results in the literature.

### 9.2.5 Generalizations

Several assumptions have been made in the development of the full information control to simplify the mathematics. Specifically, the control was assumed to be normalized so that the input-to-output coupling matrix between the control and the reference output satisfied

$$\mathbf{D}_{yu}^T \mathbf{D}_{yu} = \mathbf{I}.$$

This condition can be achieved by proper definition (normalization) of the control input. Alternatively, the  $\mathcal{H}_\infty$  suboptimal controller formulas could be modified to incorporate the use of a non-normalized control input, as given in [2], page 424. Most computer-aided design packages that allow the generation of  $\mathcal{H}_\infty$  suboptimal controllers do not require normalization of the control. In particular, normalization is not required when using the  $\mu$ -Analysis and Synthesis Toolbox of MATLAB.

The assumption that

$$\mathbf{D}_{yu}^T \mathbf{C}_y = \mathbf{0}$$

is also made in the derivation of the  $\mathcal{H}_\infty$  suboptimal controller formulas. This assumption specifies that there are no cross terms between the control and the state in the cost function. These cross terms can be incorporated into the problem statement, yielding an increase in the complexity of the  $\mathcal{H}_\infty$  suboptimal controller formulas. These more general formulas can be found in [2], page 449, and software for finding the associated controllers can be found in the  $\mu$ -Analysis and Synthesis Toolbox of MATLAB.

Terminal state weighting can be employed in the differential game used to generate the  $\mathcal{H}_\infty$  full information control:

$$J_\gamma(x, u, w) = \|y\|_2^2 - \gamma^2 \|w\|_2^2 + x^T(t_f) \mathbf{H} x(t_f).$$



The resulting controller (if it exists) is still an  $\mathcal{H}_\infty$  full information suboptimal controller. This controller can be found by solving the Riccati equation with the final condition

$$\mathbf{P}(t_f) = \mathbf{H},$$

and using this Riccati solution to find the feedback gain. Note that the final condition can influence the existence of a Riccati equation solution, in general. The use of final state weighting has a similar effect on the  $\mathcal{H}_\infty$  full information controller as on the LQR; that is, it forces the final state closer to zero.

The  $\mathcal{H}_\infty$  suboptimal full information controller is not unique. The set of all controllers that satisfy the bound can be constructed by adding terms to the given state feedback controller. A second optimization may then be performed, if desired, over this set to yield controllers that have additional properties, for example, robustness. Parameterizations for the set of all  $\mathcal{H}_\infty$  suboptimal full information controllers that satisfy a given bound can be found in [1], page 250, and [2], page 425.

### 9.3 $\mathcal{H}_\infty$ Estimation

The linear quadratic Gaussian control is generated by an optimal state feedback control law operating on estimates of the state. The states, in this case, are estimated using the Kalman filter. A similar structure exists for the  $\mathcal{H}_\infty$  output feedback controller, as presented in the next chapter. Before discussing this structure in detail, the  $\mathcal{H}_\infty$  estimation problem is posed and solved.

Two fundamental differences exist between the Kalman filter and the  $\mathcal{H}_\infty$  optimal estimator (or filter). First, the  $\mathcal{H}_\infty$  filter is optimal in terms of minimizing the  $\infty$ -norm of the gain between a set of disturbance inputs, and the estimation error. This performance criteria specifies that the worst-case gain be minimized. In contrast, the Kalman filter minimizes the mean square estimation error, or equivalently, minimizes the mean square gain between the disturbances and the estimation error.

The second difference stems from the fact that minimum mean square estimation commutes with linear operations; that is, the minimum mean square estimate of any linear combination of the state is simply the same linear combination of the optimal state estimate. The Kalman filter, which provides optimal estimates of the state, can therefore also be used to estimate any linear combination of the state. Minimal  $\infty$ -norm estimation does not possess this property, and the optimal  $\mathcal{H}_\infty$  estimator depends on the plant output being estimated.

The specification of the  $\mathcal{H}_\infty$  estimation problem requires a model of the plant and a cost function. The plant is modeled as follows:

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}_u u(t) + \mathbf{B}_w w(t); \quad (9.31a)$$

$$m(t) = \mathbf{C}_m x(t) + \mathbf{D}_{mw} w(t), \quad (9.31b)$$

where  $u(t)$  is a known input, and  $w(t)$  is an unknown (but not necessarily random) disturbance input. The plant is assumed to be observable from the measurement and

controllable from the disturbance input.<sup>8</sup> The matrices  $\mathbf{B}_w$  and  $\mathbf{D}_{mw}$  are assumed to satisfy the following conditions:

$$\mathbf{D}_{mw} \mathbf{B}_w^T = \mathbf{0}; \quad (9.31c)$$

$$\mathbf{D}_{mw} \mathbf{D}_{mw}^T = \mathbf{I}. \quad (9.31d)$$

Condition (9.31c) specifies that the disturbances entering the plant via the state equation (similar to plant noise) and the disturbances entering the plant via the measurement equation (similar to measurement noise) must be distinct. This condition is akin to requiring the measurement and plant noises be independent in the Kalman filter setting. Condition (9.31d) specifies that the output equation must be scaled to normalize the input-to-output coupling matrix between the disturbance and the measurement. This normalization can always be accomplished provided  $\mathbf{D}_{mw}$  has full rank. Assuming that the output equation is normalized is not necessary to the theory, but simplifies the subsequent derivations.

The  $\mathcal{H}_\infty$  filter estimates linear combinations of the state

$$y(t) = \mathbf{C}_y x(t), \quad (9.32)$$

given the measured output of the plant. An optimal  $\mathcal{H}_\infty$  estimator generates estimates that minimize the worst-case gain between the disturbance input and the estimation error  $e(t) = y(t) - \hat{y}(t)$ :

$$J = \|\mathbf{G}_{ew}\|_{\infty[0,t_f]} = \sup_{w \neq 0} \frac{\|y - \hat{y}\|_{2,[0,t_f]}}{\|w\|_{2,[0,t_f]}}. \quad (9.33)$$

This infinity norm can be defined over either a finite or an infinite time interval. The estimator is required to be stable when operating over an infinite time interval. The plant and estimator are shown in Figure 9.4, where the estimator is a linear system (not necessarily time-invariant) denoted  $\mathfrak{G}(\bullet)$ .

The  $\mathcal{H}_\infty$  estimation problem is solved by utilizing the duality between estimation and control. Duality was first presented to explain the similarity between the equations for the linear quadratic regulator and the Kalman filter. The following section provides a more detailed treatment of duality based on the adjoint system.

### 9.3.1 The Adjoint System

The duality between the control and estimation problems can be explained by noting that these problems are related via the adjoint operation. The adjoint system is a

<sup>8</sup>In general, the plant need only be detectable from the measurement and stabilizable from the disturbance input.



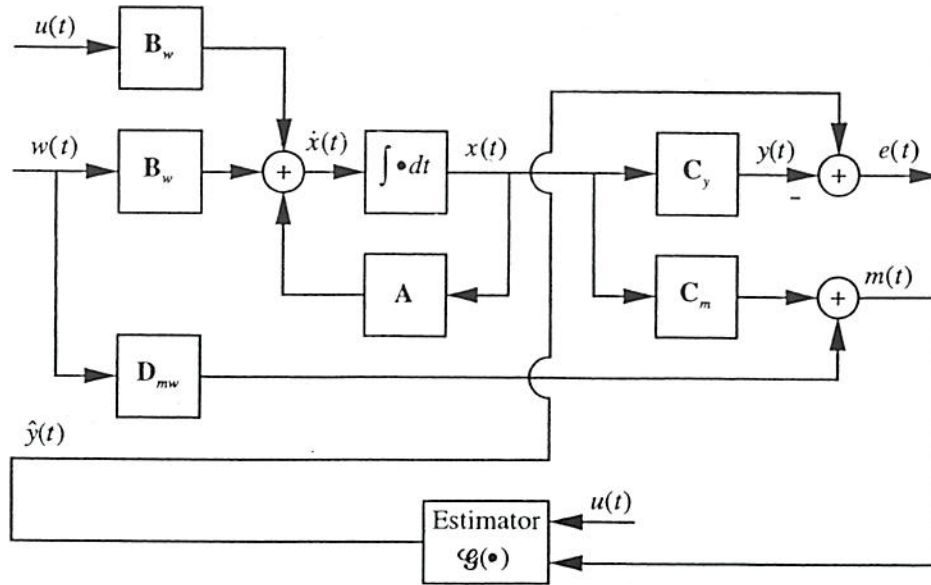


FIGURE 9.4 The estimator problem in standard form

modification of another system that has the same  $\infty$ -norm and 2-norm as the original system.<sup>9</sup>

A time-varying linear system (referred to below as the original system) is given:

$$\dot{x}(t) = \mathbf{A}(t)x(t) + \mathbf{B}(t)u(t);$$

$$y(t) = \mathbf{C}(t)x(t) + \mathbf{D}(t)u(t).$$

The adjoint system (associated with the original system) is defined as

$$\dot{\tilde{x}}(\tau) = \mathbf{A}^T(t_f - \tau)\tilde{x}(\tau)\mathbf{C}^T(t_f - \tau)\tilde{y}(\tau);$$

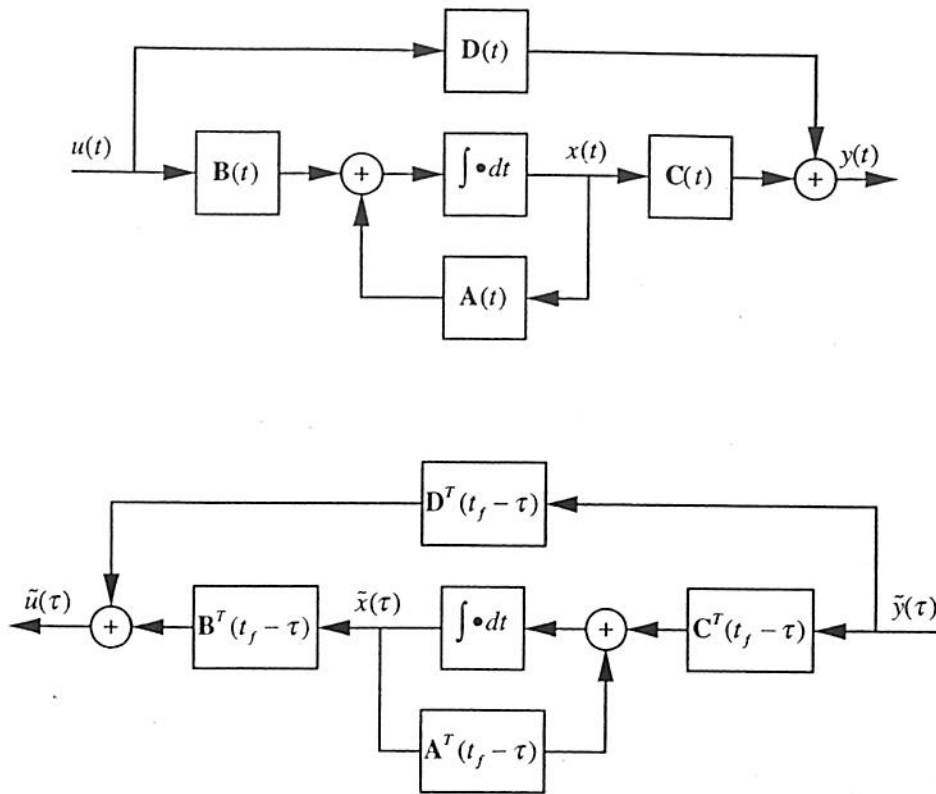
$$\tilde{u}(\tau) = \mathbf{B}^T(t_f - \tau)\tilde{x}(\tau) + \mathbf{D}^T(t_f - \tau)\tilde{y}(\tau).$$

Note that the input, output, and state of the adjoint system are distinct from the input, output, and state of the original system. Block diagrams of the original system and the adjoint system are shown in Figure 9.5. Comparing these block diagrams, note that the adjoint system is the “reverse” of the original system.<sup>10</sup> In general, the adjoint of a collection of subsystems is the adjoint of each subsystem connected in reverse order.

Two important properties of the adjoint system are that it has the same 2-norm and the same  $\infty$ -norm as the original system. These facts are rigorously demonstrated for time-varying systems in Section A11 of the Appendix. A simpler demonstration of the equivalence between the  $\infty$ -norms of the original system and the adjoint system is given

<sup>9</sup>Adjoints are typically defined by the relationship where  $\langle \mathcal{G}x, y \rangle = \langle x, \tilde{\mathcal{G}}y \rangle$ , where  $\mathcal{G}$  is a linear operator, and  $\langle \cdot, \cdot \rangle$  is an inner product. The adjoint system defined in this section is not a true adjoint, but is closely related to the true adjoint. Some authors refer to the system defined in this section as the modified adjoint system.

<sup>10</sup>The time variable in the adjoint system is typically defined as  $\tau = t_f - t$ , which reinforces the idea of the adjoint system being the reverse of the original system.



**FIGURE 9.5** Relationship between the adjoint system and the original system: (a) the original system; (b) the adjoint system

below. For this demonstration, the systems are assumed to be time-invariant, and the  $\infty$ -norm is defined over an infinite time interval.

The transfer function of a time-invariant system can be related to the transfer function of the adjoint system. The transfer function of the original system is

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} + \mathbf{D}.$$

Taking the transpose of this transfer function yields the transfer function of the adjoint system:

$$\mathbf{G}^T(s) = \mathbf{B}^T(s\mathbf{I} - \mathbf{A}^T)^{-1}\mathbf{C}^T + \mathbf{D}^T = \tilde{\mathbf{G}}(s).$$

The infinity norm of the adjoint system is then given:

$$\|\tilde{\mathbf{G}}\|_\infty = \sup_{\omega} \{\bar{\sigma}[\tilde{\mathbf{G}}(j\omega)]\} = \sup_{\omega} \{\bar{\sigma}[\mathbf{G}^T(j\omega)]\} = \sup_{\omega} \{\bar{\sigma}[\mathbf{G}(j\omega)]\} = \|\mathbf{G}\|_\infty.$$

The fact that the maximum singular value of a matrix equals the maximum singular value of the matrix transposed can be easily understood by noting that the nonzero singular values of a matrix  $\mathbf{M}$  are the nonzero eigenvalues of  $\mathbf{M}\mathbf{M}^T$ , which equal the nonzero eigenvalues of  $\mathbf{M}^T\mathbf{M}$ .



### 9.3.2 Finite-Time Estimation

The  $\mathcal{H}_\infty$  estimation problem is solved by using the adjoint to convert the estimation problem into an equivalent control problem. The full information results generated in the previous section can be used to solve this equivalent control problem. The  $\mathcal{H}_\infty$  estimator is then obtained by taking the adjoint of the resulting controller.

Paralleling the full information results, suboptimal estimators are generated that yield a given bound on the infinity norm from the disturbance input to the estimation error. General conditions can be given for the existence of these suboptimal estimators. The optimal estimator can then be approximated to an arbitrary degree of closeness by decreasing the bound until a solution no longer exists.

The equations for the plant and estimator (see Figure 9.4) are summarized as follows:

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}_w w(t); \quad (9.34a)$$

$$m(t) = \mathbf{C}_m x(t) + \mathbf{D}_{mw} w(t); \quad (9.34b)$$

$$e(t) = \mathbf{C}_y x(t) - \hat{y}(t); \quad (9.34c)$$

$$\hat{y}(t) = \mathcal{G}\{m(t)\}, \quad (9.34d)$$

where  $\mathcal{G}(\cdot)$  denotes a linear (not necessarily time-invariant) estimator. The input to the model (9.34) is  $w(t)$ , and the output is  $e(t)$ . The known input  $u(t)$  has been removed from this formulation of the estimation problem to simplify the subsequent derivations. The known input can be ignored without loss of generality, since the effects of this input on the output are easily computed. Superposition can then be invoked to add the effects of the known input into the estimator.

The adjoint of this system is obtained by taking the adjoint of the plant (9.34a) through (9.34c), taking the adjoint of the estimator (9.34d), and combining the results:

$$\dot{\tilde{x}}(\tau) = \mathbf{A}^T \tilde{x}(\tau) + \mathbf{C}_m^T \tilde{m}(\tau) + \mathbf{C}_y^T \tilde{e}(\tau); \quad (9.35a)$$

$$\tilde{w}(\tau) = \mathbf{B}_w^T \tilde{x}(\tau) + \mathbf{D}_{mw}^T \tilde{m}(\tau); \quad (9.35b)$$

$$\tilde{\hat{y}}(\tau) = -\tilde{e}(\tau); \quad (9.35c)$$

$$\tilde{m}(\tau) = \tilde{\mathcal{G}}\{\tilde{\hat{y}}(\tau)\}. \quad (9.35d)$$

A block diagram of the adjoint system is shown in Figure 9.6.

The system in Figure 9.6 is a control system. For this system, the controller  $\tilde{\mathcal{G}}\{\cdot\}$  can be selected to bound the cost:

$$J = \|\mathbf{G}_{\tilde{w}\tilde{e}}\|_{\infty, [0, t_f]} = \sup_{\|\tilde{e}\|_{2, [0, t_f]} \neq 0} \frac{\|\tilde{w}\|_{2, [0, t_f]}}{\|\tilde{e}\|_{2, [0, t_f]}} < \gamma. \quad (9.36)$$

This suboptimal control problem is very similar to the full information control problem. Note that the plant model (9.35a) and (9.35b) has exactly the form of (9.2a) and (9.2b) when  $\tilde{m}(\tau)$  is the "control input." Further, the conditions (9.30c) and (9.30d) are equivalent to (9.2c) and (9.2d) for the adjoint plant:

$$(\mathbf{D}_{mw}^T)^T \mathbf{B}_w^T = \mathbf{0};$$

$$(\mathbf{D}_{mw}^T)^T \mathbf{D}_{mw}^T = \mathbf{I}.$$

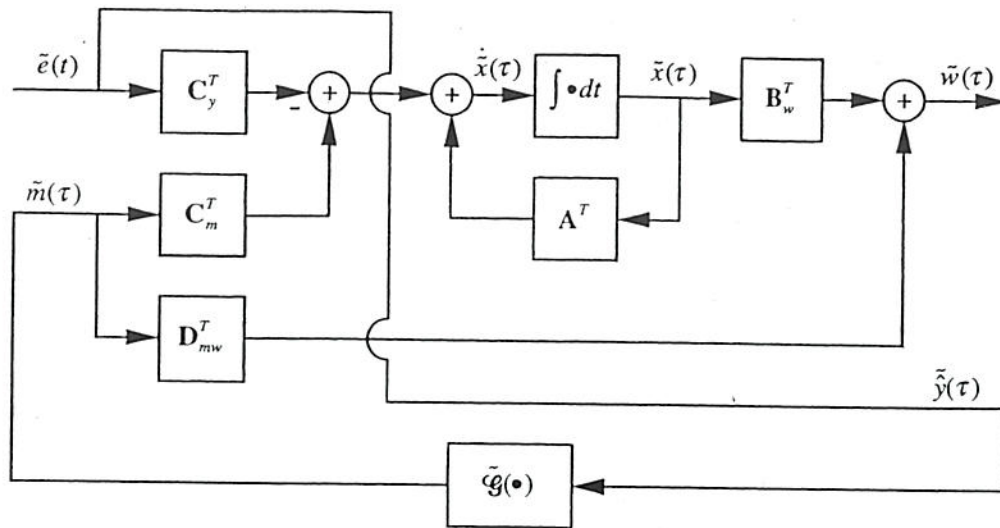


FIGURE 9.6 The adjoint of the estimator problem in standard form

Lastly, the initial condition for the adjoint system can be assumed to equal zero, since the  $\infty$ -norm only depends on the forced response.

The control problem (9.35) and (9.36) differs from the full information control problem only in the fact that the measurement available for feedback (9.35c) is the disturbance input. In contrast, both the disturbance input and the state are assumed to be available in the full information control problem. A controller that utilizes only the disturbance input is referred to as a disturbance feedforward controller.

The existence of an  $\mathcal{H}_\infty$  full information suboptimal control for the plant (9.35) is necessary for the existence of a solution to the disturbance feedforward control problem. This observation is based on the fact that a disturbance feedforward controller is a special case of a full information controller. Less obvious is the fact that the full information control problem is mathematically equivalent to the disturbance feedforward control problem.

The equivalence of the full information and disturbance feedforward control problems is demonstrated by noting that the state of a plant can be perfectly reconstructed from the initial condition and the inputs. Since the initial condition for the plant (9.35) is zero, the state can be perfectly reconstructed from the disturbance input and the control input. Since the controller always has knowledge of the control input, knowledge of the disturbance input is equivalent to full information.

The full information control results can be used to generate a suboptimal solution to the control problem in Figure 9.6. The estimator  $\mathcal{G}(\bullet)$  can then be recovered from the resulting controller  $\tilde{\mathcal{G}}(\bullet)$  by taking the adjoint. The resulting estimator is a suboptimal solution of the  $\mathcal{H}_\infty$  estimation problem, since the  $\infty$ -norm of the closed-loop adjoint system equals the  $\infty$ -norm of the original system; see Appendix equation (A10.6).

**The Riccati Equation** The disturbance feedforward controller is found by utilizing the full information results to yield a state feedback controller. The state used in this controller is reconstructed by applying the control and disturbance inputs to the adjoint model. The resulting controller is given:



$$\begin{aligned}\dot{\tilde{x}}(\tau) &= \mathbf{A}^T \tilde{x}(\tau) + \mathbf{C}_m^T \tilde{m}(\tau) + \mathbf{C}_y^T \tilde{e}(\tau); \\ \tilde{m}(\tau) &= -\tilde{\mathbf{G}}(\tau) \tilde{x}(\tau),\end{aligned}$$

or, equivalently,

$$\dot{\tilde{x}}(\tau) = [\mathbf{A}^T - \mathbf{C}_m^T \tilde{\mathbf{G}}(\tau)] \tilde{x}(\tau) + \mathbf{C}_y^T \tilde{e}(\tau); \quad (9.37a)$$

$$\tilde{m}(\tau) = \tilde{\mathbf{G}}(\tau) \tilde{x}(\tau). \quad (9.37b)$$

The state feedback gain in (9.37b) is equal to the full information control:

$$\tilde{\mathbf{G}}(\tau) = -\mathbf{C}_m \tilde{\mathbf{Q}}(\tau). \quad (9.38)$$

The matrix  $\tilde{\mathbf{Q}}(\tau)$  is found by solving the following Riccati differential equation:

$$\dot{\tilde{\mathbf{Q}}}(\tau) = -\tilde{\mathbf{Q}}(\tau) \mathbf{A}^T - \mathbf{A} \tilde{\mathbf{Q}}(\tau) - \mathbf{B}_w \mathbf{B}_w^T + \tilde{\mathbf{Q}}(\tau) (\mathbf{C}_m^T \mathbf{C}_m - \gamma^{-2} \mathbf{C}_y^T \mathbf{C}_y) \tilde{\mathbf{Q}}(\tau) \quad (9.39a)$$

backward in time from the final condition

$$\tilde{\mathbf{Q}}(t_f) = \mathbf{0}. \quad (9.39b)$$

The  $\mathcal{H}_\infty$  suboptimal estimator is the adjoint of the controller (9.37):

$$\begin{aligned}\dot{\hat{x}}(t) &= [\mathbf{A} - \tilde{\mathbf{G}}^T(t_f - t) \mathbf{C}_m] \hat{x}(t) + \tilde{\mathbf{G}}^T(t_f - t) m(t); \\ e(t) &= \mathbf{C}_y \hat{x}(t).\end{aligned}$$

Defining,

$$\mathbf{G}(t) = \tilde{\mathbf{G}}^T(t_f - t), \quad (9.40)$$

adding the known input into the estimator equation, and rearranging yields the  $\mathcal{H}_\infty$  suboptimal estimator:

$$\dot{\hat{x}}(t) = \mathbf{A} \hat{x}(t) + \mathbf{B}_u u(t) + \mathbf{G}(t) \{m(t) - \mathbf{C}_y \hat{x}(t)\}; \quad (9.41a)$$

$$\hat{y}(t) = \mathbf{C}_y \hat{x}(t). \quad (9.41b)$$

A block diagram of this estimator is shown in Figure 9.7. Note that this estimator has the structure of the Kalman filter, but the estimator gain is selected to minimize the  $\infty$ -norm criterion.

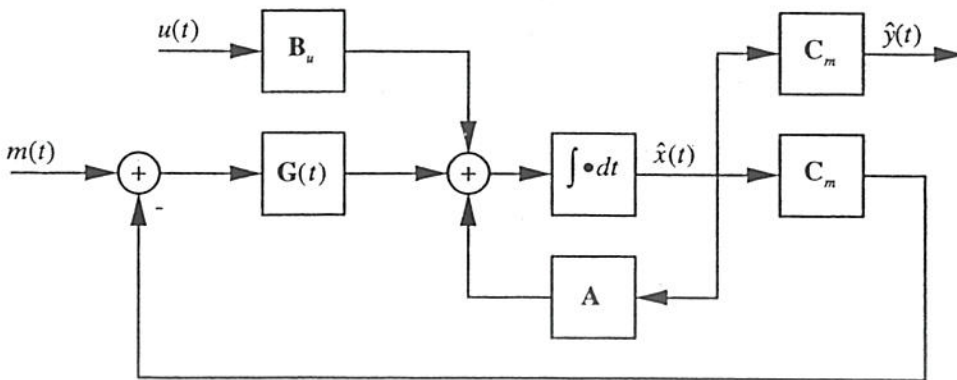


FIGURE 9.7 The  $\mathcal{H}_\infty$  optimal estimator

The Riccati equation (9.39) can be placed in a more convenient form by performing the change of variables  $\tau = t_f - t$ , and defining

$$\mathbf{Q}(t) = \tilde{\mathbf{Q}}(t_f - \tau).$$

A Riccati equation for this new matrix is then

$$\dot{\tilde{\mathbf{Q}}}(\tau) = \tilde{\mathbf{Q}}(\tau)\mathbf{A}^T + \mathbf{A}\tilde{\mathbf{Q}}(\tau) + \mathbf{B}_w\mathbf{B}_w^T - \tilde{\mathbf{Q}}(\tau)(\mathbf{C}_m^T\mathbf{C}_m - \gamma^{-2}\mathbf{C}_y^T\mathbf{C}_y)\tilde{\mathbf{Q}}(\tau).$$

(9.42a)

This Riccati equation is solved forward in time from the initial condition

$$\tilde{\mathbf{Q}}(0) = \mathbf{0}.$$

(9.42b)

The estimator gain can be written in terms of this new Riccati solution:

$$\mathbf{G}(t) = \tilde{\mathbf{Q}}(t)\mathbf{C}_m^T.$$

(9.43)

Equations (9.41) through (9.43) completely specify the suboptimal  $\mathcal{H}_\infty$  estimator. Note that the Riccati equation and therefore the estimator gain depend on the output being estimated, since this equation contains  $\mathbf{C}_y$ .

### EXAMPLE 9.3

An  $\mathcal{H}_\infty$  estimator can be applied to the radar range tracking of an aircraft (first presented in Example 7.2). A state equation for the range  $r(t)$  and radial velocity of an aircraft is

$$\begin{bmatrix} \dot{r}(t) \\ \dot{\dot{r}}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w(t).$$

The range measurements are given,

$$m(t) = [1 \quad 0] \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix} + v(t),$$

and the output to be estimated is the entire state:

$$y(t) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix}.$$

The plant input and measurement error are assumed to be bounded:

$$\begin{bmatrix} |w(t)| \\ |v(t)| \end{bmatrix} \leq \begin{bmatrix} 4 \text{ m/sec}^2 \\ 20 \text{ m} \end{bmatrix}.$$

A state equation of the form in (9.31a) can be generated by including the measurement error as part of the disturbance input:

$$\begin{bmatrix} \dot{r}(t) \\ \dot{\dot{r}}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} w_1(t) \\ v_1(t) \end{bmatrix}.$$



The inputs are normalized, and the plant input bound is included as a weight in this state equation. Appending the measurement error bound to the measurement equation yields

$$m(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix} + \begin{bmatrix} 0 & 20 \end{bmatrix} \begin{bmatrix} w_1(t) \\ v_1(t) \end{bmatrix}.$$

Normalizing this equation so that the input-to-output coupling matrix satisfies the constraint (9.31d) gives

$$m_1(t) = \begin{bmatrix} \frac{1}{20} & 0 \end{bmatrix} \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix} + \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} w_1(t) \\ v_1(t) \end{bmatrix},$$

where  $m(t) = 20m_1(t)$ . This measurement equation and the state equation (including the input bound) are in a form appropriate for applying (9.41) through (9.43) for  $\mathcal{H}_\infty$  estimator synthesis.

The resulting  $\mathcal{H}_\infty$  estimator is given as

$$\begin{aligned} \dot{\hat{x}}(t) &= \mathbf{A}\hat{x}(t) + \mathbf{G}_1(t)\{m_1(t) - \begin{bmatrix} \frac{1}{20} & 0 \end{bmatrix}\hat{x}(t)\}; \\ \hat{y}(t) &= \mathbf{C}_y\hat{x}(t). \end{aligned}$$

The state equation for this estimator can be written in terms of the original measurement:

$$\begin{aligned} \dot{\hat{x}}(t) &= \mathbf{A}\hat{x}(t) + \frac{1}{20}\mathbf{G}_1(t)\{20m_1(t) - 20\begin{bmatrix} \frac{1}{20} & 0 \end{bmatrix}\hat{x}(t)\} \\ &= \mathbf{A}\hat{x}(t) + \mathbf{G}(t)\{m(t) - \begin{bmatrix} 1 & 0 \end{bmatrix}\hat{x}(t)\}, \end{aligned}$$

where  $\mathbf{G}(t) = \mathbf{G}_1(t)/20$ .<sup>11</sup>

The plant and the  $\mathcal{H}_\infty$  estimator are simulated. The disturbance input and the measurement error are both assumed to be discrete-time white noise, uniformly distributed within the given bounds. An initial estimation error is included in the simulation to display the transient performance of the estimator, even though this initial error is assumed to be zero during estimator development.

The estimator gains and the estimated outputs (the states in this case) are shown in Figure 9.8, where the estimator performance bound is:  $\gamma = 22.5$ . This particular performance bound is selected because it is roughly 10% larger than the optimal bound.<sup>12</sup> The gains converge to steady-state values. Also, the estimates converge in slightly over 10 seconds for both position and velocity, and the estimator tracks the actual state. Figure 9.9 shows the estimator gains and the estimated output when the bound on the measurement error is decreased to 2. In this case, an estimator performance bound of  $\gamma = 3.5$  ( $\sim 10\%$  over optimal) is used. The estimates are seen to converge in less than

<sup>11</sup> The normalization of the measurement equation and conversion of the estimator state equation to utilize the original measurements can typically be handled by CAD software packages. For example, normalization of the measurement is not required when using the  $\mu$ -Synthesis and Analysis Toolbox of MATLAB.

<sup>12</sup> Numerical difficulties are frequently encountered when doing discrete-time implementations of near-optimal  $\mathcal{H}_\infty$  estimators. These difficulties are mostly avoided by making the bound slightly larger than optimal.

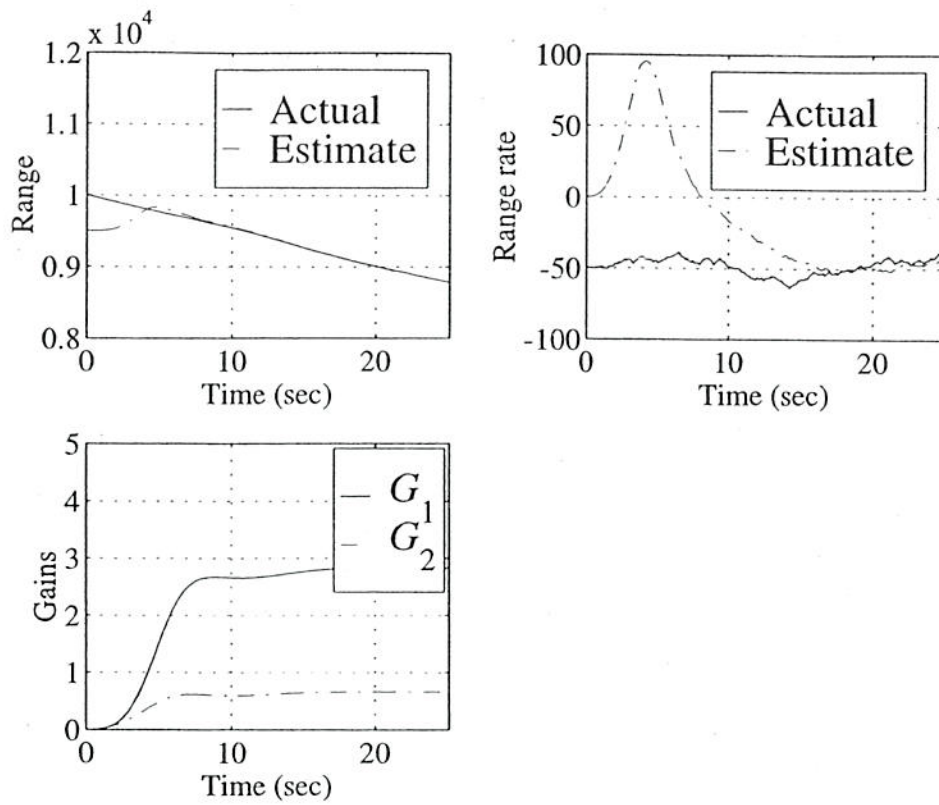


FIGURE 9.8 Baseline results for Example 9.3

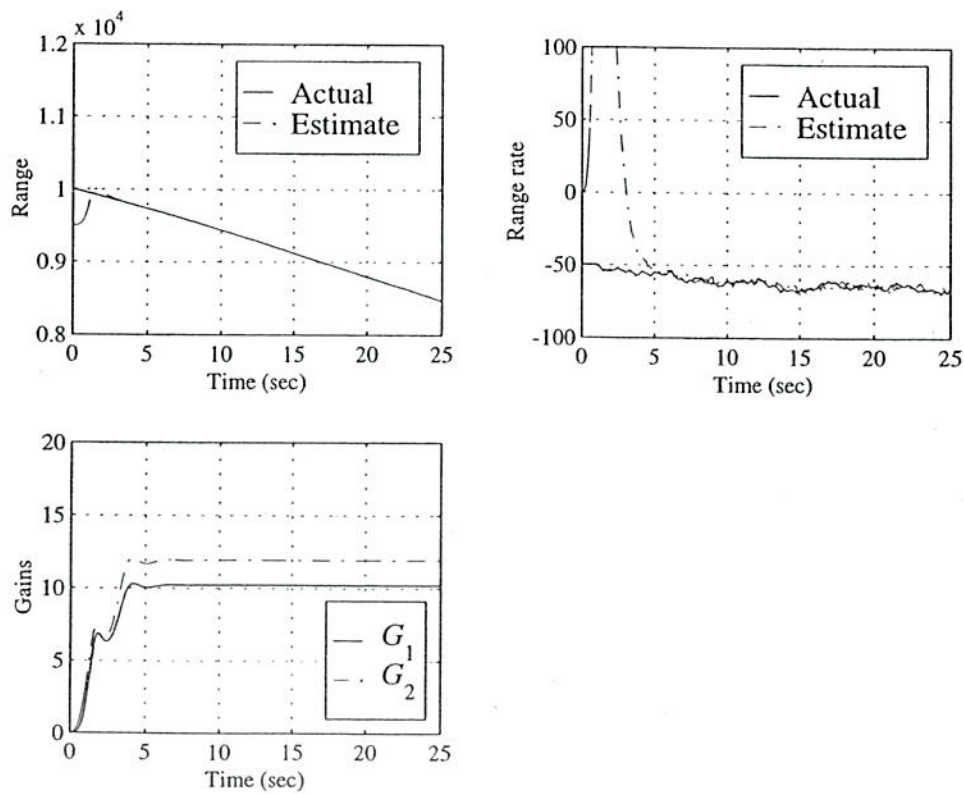


FIGURE 9.9 Results for Example 9.3 with decreased measurement error



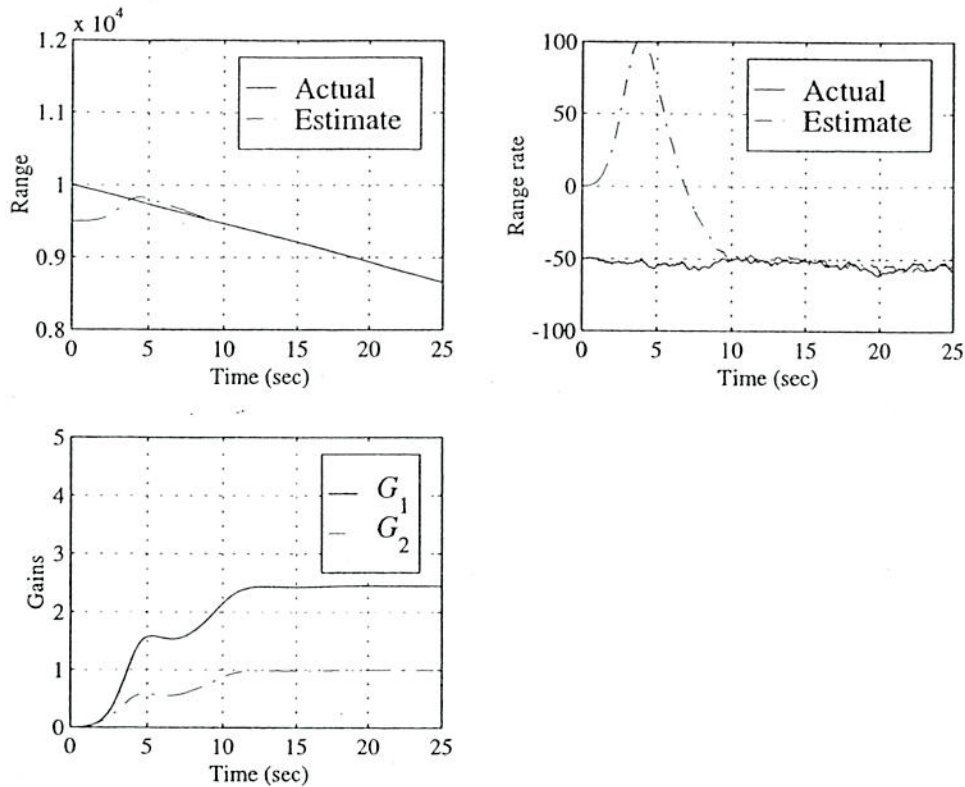


FIGURE 9.10 Results for Example 9.3 with only range rate estimation

5 seconds, about twice as fast as in the baseline case. This faster rate of convergence results from the larger estimator gains, and also increases the overshoot in the velocity estimate. This is expected from the Kalman filtering results, where faster convergence occurs when the spectral density of the measurement noise is decreased.

Figure 9.10 shows the estimator gains and the estimated states when the reference output is only the range rate:

$$y(t) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix}.$$

Note both range and range rate are still available as the estimator states and are therefore included in the figure. In this case, an estimator performance bound of  $\gamma = 10$  ( $\sim 10\%$  over optimal) was used. When exclusively estimating range rate, the gains change slightly. Again, it is worth noting that the  $\mathcal{H}_\infty$  estimator depends on the output of interest. ♦

The Riccati equation (9.42) approaches the Kalman Riccati equation as the performance bound  $\gamma$  approaches infinity.<sup>13</sup> A solution is therefore guaranteed to exist for

<sup>13</sup>The covariance matrices of the plant driving noise and the measurement noise are assumed to both equal identity matrices to yield the associated Kalman filtering problem. This can be accomplished, in general, by absorbing these covariance matrices into the input matrix for the plant noise, and also by appropriately scaling the measurement equation (normalizing the measurement noise).

sufficiently large performance bounds. For finite performance bounds, this equation may or may not have a solution. In general, solutions exist for performance bounds above a limiting value equal to the optimal value, where optimal is defined in terms of minimizing the norm (9.33).

**The Hamiltonian Equations** The suboptimal full information controller, and therefore the suboptimal estimator, can be generated by solving a Riccati equation or by solving the Hamiltonian equations. The Hamiltonian equations for the estimator can be combined to form the Hamiltonian system:

$$\begin{bmatrix} \dot{\tilde{x}}(t) \\ \dot{\tilde{p}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{A}^T & \vdots & -\mathbf{C}_m^T \mathbf{C}_m + \gamma^{-2} \mathbf{C}_y^T \mathbf{C}_y \\ -\mathbf{B}_w^T \mathbf{B}_w^T & \vdots & -\mathbf{A} \end{bmatrix} \begin{bmatrix} \tilde{x}(t) \\ \tilde{p}(t) \end{bmatrix} = \mathbf{y}_\infty \begin{bmatrix} \tilde{x}(t) \\ \tilde{p}(t) \end{bmatrix}, \quad (9.44)$$

where the matrix  $\mathbf{y}_\infty$  is called the Hamiltonian.

The  $\mathcal{H}_\infty$  suboptimal estimator gain can be found by solving the Hamiltonian system subject to the final condition

$$\tilde{p}(t_f) = 0.$$

This gain is then given as

$$\mathbf{G}(t) = -\mathbf{Q}(t) \mathbf{C}_m^T = -\Phi_{21}^T(t) \{\Phi_{22}(t)\}^{-T} \mathbf{C}_m^T = -\{\Phi_{22}(t)\}^{-1} \Phi_{21}(t) \mathbf{C}_m^T, \quad (9.45)$$

where the terms  $\Phi_{ij}$  are blocks in the state-transition matrix of the Hamiltonian system:

$$e^{\mathbf{y}_\infty t} = \begin{bmatrix} \Phi_{11}(t) & \vdots & \Phi_{12}(t) \\ \Phi_{21}(t) & \vdots & \Phi_{22}(t) \end{bmatrix}.$$

### 9.3.3 Steady-State Estimation

The  $\mathcal{H}_\infty$  suboptimal estimator becomes time-invariant far from the initial time provided the bound is chosen sufficiently large that a steady-state solution to the Riccati equation exists. When operating over long time intervals, it is often desirable to use the steady-state estimator to simplify implementation. Mathematically, the steady-state estimator must be stable for the  $\infty$ -norm (9.33) to be defined over infinite time intervals. In practice, stable estimators are also desirable for operation over moderate to long time intervals, especially when initial errors and unmodeled dynamics are present. Thus the steady-state  $\mathcal{H}_\infty$  suboptimal estimator problem is to find a stable, time-invariant estimator that satisfies an  $\infty$ -norm (defined over an infinite time interval) bound on the transfer function from the disturbance input to the estimation error.

Conditions for the existence of an  $\mathcal{H}_\infty$  suboptimal estimator are summarized below. When these conditions are satisfied, the steady-state estimator is given by (9.41), with the gain found by using the steady-state Riccati solution in (9.42). These existence results and estimator equations are given without proof, since they can be related to the full information results via duality.



An estimator that is stable and bounds the  $\infty$ -norm of the closed-loop transfer function (9.33) exists if and only if there exists a positive semidefinite solution of the algebraic Riccati equation

$$\mathbf{A}\mathbf{Q} + \mathbf{Q}\mathbf{A}^T - \mathbf{Q}(\mathbf{C}_m^T \mathbf{C}_m - \gamma^{-2} \mathbf{C}_y^T \mathbf{C}_y) \mathbf{Q} + \mathbf{B}_w \mathbf{B}_w^T = \mathbf{0}, \quad (9.46)$$

such that the matrix

$$\mathbf{A} - \mathbf{Q}(\mathbf{C}_m^T \mathbf{C}_m - \gamma^{-2} \mathbf{C}_y^T \mathbf{C}_y) \quad (9.47)$$

is stable.

The steady-state Riccati solution can be found from the eigensolution of the Hamiltonian:

$$\mathbf{Q} = \mathbf{\Psi}_{21}(\mathbf{\Psi}_{21})^{-1},$$

where

$$\begin{bmatrix} \mathbf{\Psi}_{11} \\ \hline \mathbf{\Psi}_{21} \end{bmatrix}$$

is a matrix whose columns are the eigenvectors of the Hamiltonian associated with the stable eigenvalues. As observed previously, the  $\mathcal{H}_\infty$  optimal estimator can be approximated to an arbitrary degree of closeness by iteration of the bound and testing for the existence of suboptimal estimators.

**XAMPLE 9.4** An ac motor is described by the following state equation:

$$\dot{x}(t) = -x(t) + 100u(t) + 100w(t),$$

where  $x(t)$  is the rotational velocity of the shaft,  $u(t)$  is the nominal input of 10 volts rms, and  $w(t)$  is the error between the actual applied voltage and the nominal value. This error is assumed to be caused by modulation of the ac envelope due to harmonics on the power line. This modulation is assumed to be less than 1%, implying the disturbance input can be bounded:

$$|w(t)| \leq 0.1 \text{ volts rms.}$$

An  $\mathcal{H}_\infty$  estimator is used to estimate the true rotational velocity of the shaft from the noisy tachometer measurements:

$$m(t) = x(t) + v(t),$$

where  $v(t)$  is the measurement error. This error consists of a possible dc bias and sinusoidal interference, and is bounded:

$$|v(t)| \leq 1 \text{ volts rms.}$$

The output of interest is simply the state.

The Hamiltonian for this estimation problem, with the bound on the disturbance input appended to the plant as a weight, is

$$\mathcal{H}_\infty = \begin{bmatrix} -1 & -1 + \gamma^{-2} \\ -100 & 1 \end{bmatrix}.$$

Letting  $\gamma = 1.1$ , the eigenvalues and eigenvectors of the Hamiltonian are

$$\lambda_1 = -4.28 ; \lambda_2 = 4.28 ; \phi_1 = \begin{bmatrix} -0.0528 \\ -0.9986 \end{bmatrix} ; \phi_2 = \begin{bmatrix} 0.0328 \\ -0.9995 \end{bmatrix}.$$

The resulting normalized estimator gain is

$$G_1 = 18.92,$$

yielding the estimator

$$\begin{aligned} \dot{\hat{x}}(t) &= -\hat{x}(t) + 100u(t) + 1.892\{m(t) - \hat{x}(t)\} \\ &= -2.892\hat{x}(t) + 100u(t) + 1.892m(t). \end{aligned}$$

The output of interest is the state in this example, so no output equation is required.

The plant and the  $\mathcal{H}_\infty$  estimator are simulated. The disturbance input is a 2 Hz sinusoid of amplitude 0.1. The measurement error is a dc bias with an amplitude of 1. The initial state and initial estimate are both set equal to 1000 rad/sec, which is the nominal motor shaft velocity, since only steady-state performance is of interest. The plant state and the estimated state are shown in Figure 9.11. Note that the maximum error exceeds the performance bound. This is reasonable since the performance bound

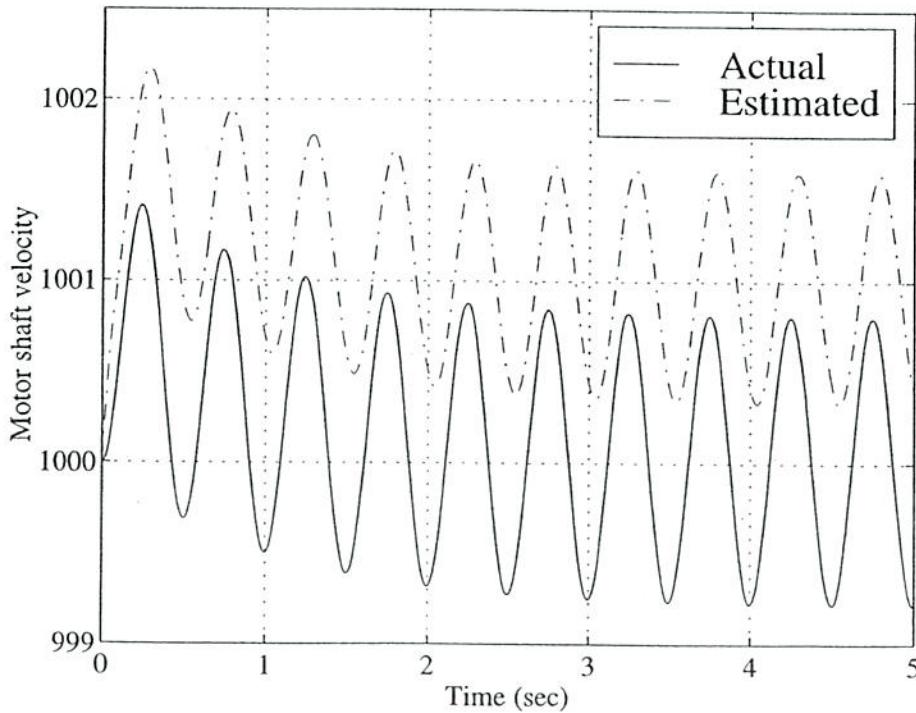


FIGURE 9.11  $\mathcal{H}_\infty$  estimation of motor shaft velocity



assumes that both inputs have the same frequency and the sum of the squares of their amplitudes is bounded by 1 (for the normalized state equation). In reality, the inputs have distinct frequencies and each achieves the maximum amplitude. Still, the performance bound provides a good indication of the observed estimation error. ♦

### 9.3.4 Generalizations

Normalizing the measurement error simplifies the formulas for the  $\mathcal{H}_\infty$  suboptimal estimator. More general formulas can be derived that do not require this normalization. Generalizations also exist to the estimator formulas, which allow a particular disturbance input to enter both the state equation and the measurement equation. This generalization is equivalent to allowing correlation between the plant and measurement noises in the Kalman filter problem. These more general  $\mathcal{H}_\infty$  suboptimal estimator formulas are derived in [2], page 449, and software for finding the associated estimators can be found in the  $\mu$ -Analysis and Synthesis Toolbox of MATLAB.

The initial condition on the Riccati equation can be modified to reflect uncertainty in the initial state, with a large initial Riccati solution being used when the state is very uncertain. The initial condition on the Riccati equation has an effect on the  $\mathcal{H}_\infty$  estimator that is similar to the effect of the initial condition covariance matrix on the Kalman filter. In fact, it is reasonable to use the initial condition covariance matrix when initializing the  $\mathcal{H}_\infty$  Riccati equation. The resulting estimator (if it exists) is still an  $\mathcal{H}_\infty$  suboptimal estimator, but the initial condition can influence the existence of a Riccati equation solution.

A parameterization exists for the set of all possible  $\mathcal{H}_\infty$  suboptimal estimators that satisfy a given performance bound; see [1], page 266, and [2], page 433. This parameterization allows the designer to perform a second optimization to maximize or minimize an additional performance criteria. Note that the resulting estimator still satisfies the original estimation error bound.

Smoothing involves estimating the output based on both past and future measurements. While smoothing cannot be applied in real time, it is useful for off-line data analysis.  $\mathcal{H}_\infty$  smoothing algorithms can be developed on the basis of the  $\mathcal{H}_\infty$  output estimation results given in this chapter.  $\mathcal{H}_\infty$  smoothing is presented in [3].

## 9.4 Summary

The  $\mathcal{H}_\infty$  optimal full information control problem is to find a feedback controller that utilizes the plant state and the disturbance input, and minimizes the closed-loop system  $\infty$ -norm. This feedback controller is also required to internally stabilize the system in the infinite-time case. Differential game theory is used to generate a solution to the full information problem. Unfortunately, differential game theory does not directly yield a solution to the optimal full information problem. Instead, a suboptimal controller is generated that satisfies a bound on the closed-loop system  $\infty$ -norm. This suboptimal controller is state feedback with a gain that is given in terms of the solution of a Hamiltonian equation or, equivalently, in terms of the solution of a Riccati equation. General conditions are given for the existence of a suboptimal controller that satisfies a given  $\infty$ -norm bound. Therefore, the optimal solution can be approximated arbitrarily closely by decreasing the bound until a suboptimal controller no longer exists.



A suboptimal, steady-state full information controller is generated as a limit of the finite-time full information controller. The bounded real lemma is used to demonstrate that this controller satisfies the desired  $\infty$ -norm bound. The existence results for the steady-state full information controller are given both in terms of a Riccati equation and in terms of a Hamiltonian equation. In both cases, an additional condition is required to guarantee internal stability of the feedback system.

The  $\mathcal{H}_\infty$  estimation problem is to estimate a linear combination of the plant state, given measurements and known plant inputs. The resulting estimator should minimize the  $\infty$ -norm of the transfer function between the disturbance input (including measurement error) and the estimation error. This estimator is also required to be stable in the infinite-time case. The  $\mathcal{H}_\infty$  estimator, while technically not the dual of the full information controller, is still derived using the full information results and duality arguments. The resulting suboptimal estimators have the observer structure of the Kalman filter. The observer gain is given in terms of the solution of either a Riccati equation or a Hamiltonian equation. This gain is dependent on the output being estimated, which is a fundamental distinction between  $\mathcal{H}_\infty$  estimation and Kalman filtering. General conditions that guarantee the existence of an  $\mathcal{H}_\infty$  suboptimal estimator are also given. The optimal estimator can be approximated by decreasing the bound on the  $\infty$ -norm until a solution no longer exists.

Game theory can be applied directly to the discrete-time plant model to yield a suboptimal,  $\mathcal{H}_\infty$  full information controller [4]. An alternative approach to discrete-time  $\mathcal{H}_\infty$  controller synthesis is presented in [5]. The fundamental equations for the steady-state  $\mathcal{H}_\infty$  full information controller and the steady-state  $\mathcal{H}_\infty$  estimator are summarized for both continuous-time and discrete-time systems in Table 9.1.

**TABLE 9.1** Summary of Formulas for Both Continuous-Time and Discrete-Time Systems

| Formula                                                  | Continuous-Time                                                                                                                                                                                                                                                                                                                                                                                                  | Discrete-Time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ie $\mathcal{H}_\infty$ full information control problem | Given the plant<br>$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}_u u(t) + \mathbf{B}_w w(t);$<br>$y(t) = \mathbf{C}_y x(t) + \mathbf{D}_{yu} u(t),$<br>where $\mathbf{D}_{yu}^T \mathbf{C}_y = \mathbf{0}; \mathbf{D}_{yu}^T \mathbf{D}_{yu} = \mathbf{I}$ , generate a feedback controller such that<br>$\ \mathbf{G}_{yw}\ _\infty < \gamma.$                                                                      | Given the plant<br>$\dot{x}(t) = \Phi x(t) + \Gamma_u u(t) + \Gamma_w w(t);$<br>$y(t) = \mathbf{C}_y x(t) + \mathbf{D}_{yu} u(t),$<br>where $\mathbf{D}_{yu}^T \mathbf{C}_y = \mathbf{0}; \mathbf{D}_{yu}^T \mathbf{D}_{yu} = \mathbf{I}$ , generate a feedback controller such that <sup>1</sup><br>$\ \mathbf{G}_{yw}\ _\infty < \gamma.$                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| existence conditions for a suboptimal control            | There exists a solution $\mathbf{P} \geq \mathbf{0}$ of the algebraic Riccati equation<br>$\mathbf{P}\mathbf{A} + \mathbf{A}^T \mathbf{P} - \mathbf{P}(\mathbf{B}_u \mathbf{B}_u^T - \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T) \mathbf{P}$<br>$+ \mathbf{C}_y^T \mathbf{C}_y = \mathbf{0},$<br>where $\mathbf{A} - (\mathbf{B}_u \mathbf{B}_u^T - \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^T) \mathbf{P}$ is stable. | There exists a solution $\mathbf{P} \geq \mathbf{0}$ of the algebraic Riccati equation<br>$\mathbf{P} = \mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{A}^T \mathbf{P} [\mathbf{B}_w \ \mathbf{B}_u] \mathbf{R}^{-1} \begin{bmatrix} \mathbf{B}_w^T \\ \mathbf{B}_u^T \end{bmatrix} \mathbf{P} \mathbf{A}$<br>$+ \mathbf{C}_y^T \mathbf{C}_y,$<br>where $\mathbf{A} - [\mathbf{B}_w \ \mathbf{B}_u] \mathbf{R}^{-1} \begin{bmatrix} \mathbf{B}_w^T \\ \mathbf{B}_u^T \end{bmatrix} \mathbf{P} \mathbf{A}$ is stable, and<br>$\mathbf{R} = \begin{bmatrix} -\gamma^2 \mathbf{I} + \mathbf{B}_w^T \mathbf{P} \mathbf{B}_w & \mathbf{B}_w^T \mathbf{P} \mathbf{B}_u \\ \mathbf{B}_u^T \mathbf{P} \mathbf{B}_w & \mathbf{I} + \mathbf{B}_u^T \mathbf{P} \mathbf{B}_u \end{bmatrix}.$ |
| ie suboptimal control                                    | $u(t) = -\mathbf{B}_u^T \mathbf{P} x(t) = -\mathbf{K} x(t)$                                                                                                                                                                                                                                                                                                                                                      | $u(k) = -(\mathbf{I} + \mathbf{B}_u^T \mathbf{P} \mathbf{B}_u)^{-1} \mathbf{B}_u^T \mathbf{P} \mathbf{A} x(k)$<br>$-(\mathbf{I} + \mathbf{B}_u^T \mathbf{P} \mathbf{B}_u)^{-1} \mathbf{B}_u^T \mathbf{P} \mathbf{B}_w w(k)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

continued



TABLE 9.1 *Continued*

| Formula                                         | Continuous-Time                                                                                                                                                                                                                                                                                                                                                                                                                         | Discrete-Time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The $\mathcal{H}_\infty$ estimation problem     | <p>Given the plant</p> $\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}_u u(t) + \mathbf{B}_w w(t);$ $m(t) = \mathbf{C}_m x(t) + \mathbf{D}_{mw} w(t),$ $Y(t) = \mathbf{C}_y x(t)$ <p>where <math>\mathbf{D}_{mw} \mathbf{B}_w^T = \mathbf{0}</math>; <math>\mathbf{D}_{mw} \mathbf{D}_{mw}^T = \mathbf{I}</math>, generate an estimator such that</p> $\ \mathbf{G}_{ew}\ _\infty < \gamma.$                                                  | <p>Given the plant</p> $\dot{x}(t) = \Phi x(t) + \Gamma_u u(t) + \Gamma_w w(t);$ $m(t) = \mathbf{C}_m x(t) + \mathbf{D}_{mw} w(t),$ $Y(t) = \mathbf{C}_y x(t)$ <p>where <math>\mathbf{D}_{mw} \mathbf{B}_w^T = \mathbf{0}</math>; <math>\mathbf{D}_{mw} \mathbf{D}_{mw}^T = \mathbf{I}</math>, generate an estimator such that</p> $\ \mathbf{G}_{ew}\ _\infty < \gamma.$                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Existence conditions for a suboptimal estimator | <p>There exists a solution <math>\mathbf{Q} \geq \mathbf{0}</math> of the algebraic Riccati equation</p> $\mathbf{A}\mathbf{Q} + \mathbf{Q}\mathbf{A}^T - \mathbf{Q}(\mathbf{C}_m^T \mathbf{C}_m - \gamma^{-2} \mathbf{C}_y^T \mathbf{C}_y) \mathbf{Q} + \mathbf{B}_w \mathbf{B}_w^T = \mathbf{0},$ <p>where <math>\mathbf{A} - \mathbf{Q}(\mathbf{C}_m^T \mathbf{C}_m - \gamma^{-2} \mathbf{C}_y^T \mathbf{C}_y)</math> is stable.</p> | <p>There exists a solution <math>\mathbf{Q} \geq \mathbf{0}</math> of the algebraic Riccati equation</p> $\mathbf{Q} = \mathbf{A}\mathbf{Q}\mathbf{A}^T - \mathbf{A}\mathbf{Q}[\mathbf{C}_y^T \mid \mathbf{C}_m^T] \mathbf{S}^{-1} \begin{bmatrix} \mathbf{C}_y \\ \mathbf{C}_m \end{bmatrix} \mathbf{Q}\mathbf{A}^T + \mathbf{B}_w \mathbf{B}_w^T,$ <p>where <math>\mathbf{A} - \mathbf{A}\mathbf{Q}[\mathbf{C}_y^T \mid \mathbf{C}_m^T] \mathbf{S}^{-1} \begin{bmatrix} \mathbf{C}_y \\ \mathbf{C}_m \end{bmatrix}</math> is stable, and</p> $\mathbf{S} = \begin{bmatrix} -\gamma^2 \mathbf{I} + \mathbf{C}_y \mathbf{Q} \mathbf{C}_y^T & \mathbf{C}_y \mathbf{Q} \mathbf{C}_m^T \\ \mathbf{C}_m \mathbf{Q} \mathbf{C}_y^T & \mathbf{I} + \mathbf{C}_m \mathbf{Q} \mathbf{C}_m^T \end{bmatrix}.$ |
| The suboptimal estimator                        | $\dot{\hat{x}}(t) = \mathbf{A}\hat{x}(t) + \mathbf{B}_u u(t) + \mathbf{G}\{m(t) - \mathbf{C}_m \hat{x}(t)\}$ $\hat{y}(t) = \mathbf{C}_y \hat{x}(t)$ $\mathbf{G} = \mathbf{Q}\mathbf{C}_m^T$                                                                                                                                                                                                                                             | $\hat{x}(k+1) = \Phi \hat{x}(k) + \Gamma_u u(k) + \mathbf{G}\{m(k) - \mathbf{C}_m \hat{x}(k)\}$ $\hat{y}(t) = \mathbf{C}_y \hat{x}(t) + \mathbf{G}_r \{m(t) - \mathbf{C}_m \hat{x}(t)\}$ $\mathbf{G} = \mathbf{A}\mathbf{Q}\mathbf{C}_m^T (\mathbf{I} + \mathbf{C}_m \mathbf{Q} \mathbf{C}_m^T)^{-1}$ $\mathbf{G}_r = \mathbf{C}_y \mathbf{Q} \mathbf{C}_m^T (\mathbf{I} + \mathbf{C}_m \mathbf{Q} \mathbf{C}_m^T)^{-1}$                                                                                                                                                                                                                                                                                                                                                                            |

<sup>1</sup> For discrete-time systems, the signal 2-norm and the system  $\infty$ -norm are defined as

$$\|w(k)\|_2 = \left\{ \sum_{k=-\infty}^{\infty} w^T(k)w(k) \right\}^{1/2}; \quad \|\mathbf{G}\|_\infty = \sup_{w(k) \neq 0} \frac{\|\mathbf{g}(k) \otimes w(k)\|_2}{\|w(k)\|_2}.$$

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